🚙 Smart Vehicle Monitoring & Anti-Theft System

Detailed Implementation Guide

📌 1. Project Overview

This project is a **low-cost embedded anti-theft system** that monitors a vehicle for suspicious activity.

The system uses:

- 📳 **Vibration Sensor** → Detects tampering/shaking
- 📐 **Tilt Sensor** → Detects lifting or towing
- 📺 **OLED Display** → Shows status/alerts
- 🔊 **Buzzer** → Generates alarm

When abnormal activity is detected:

- Alarm is activated
- OLED displays warning
- Serial monitor logs event

🎯 2. Objectives

- Detect unauthorized access
- Detect movement/lifting of vehicle
- Trigger real-time alerts
- Build a scalable IoT-ready system

🧩 3. Components Required

✅ Main Components

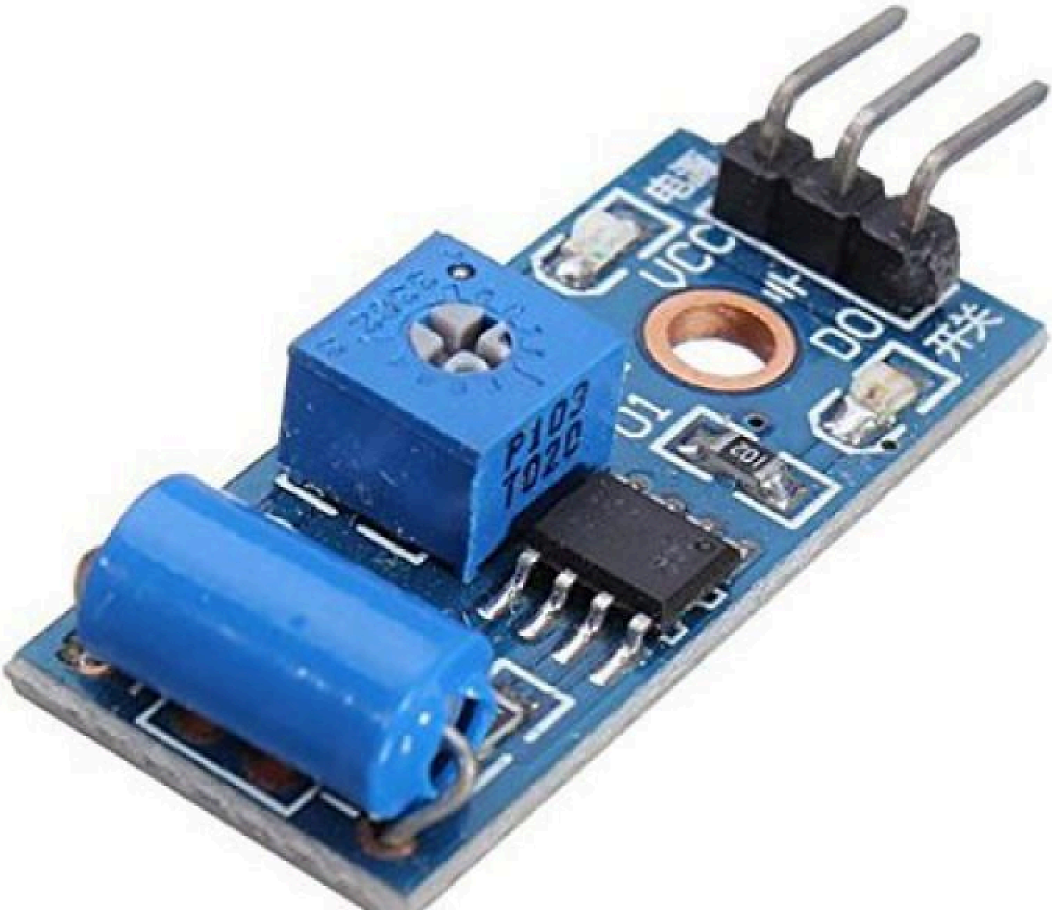
Component	Quantity	Purpose
Arduino Uno	1	Main controller
OLED Display (SSD1306 I2C 128×64)	1	Display alerts/status
SW-420 Vibration Sensor	1	Detects vibration
Tilt Sensor	1	Detects angle change
Active Buzzer	1	Alarm sound
Jumper Wires	~15	Circuit connections
USB Cable	1	Programming & power

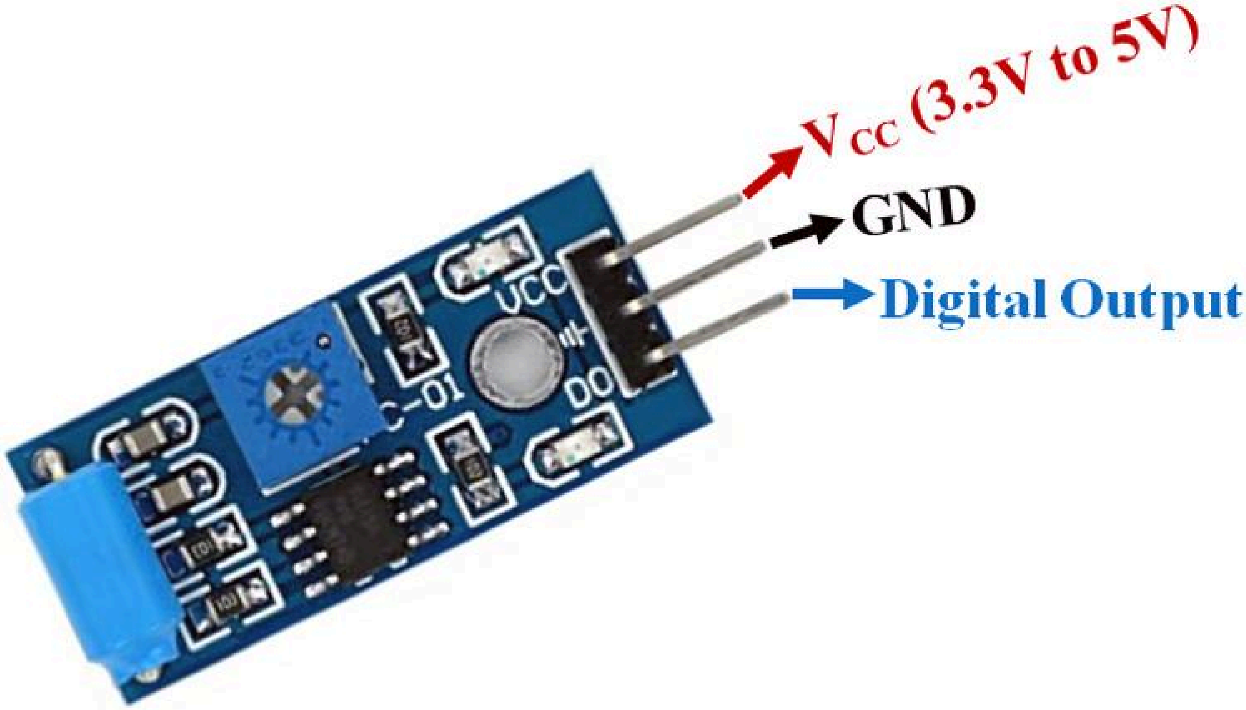
➕ Optional Components

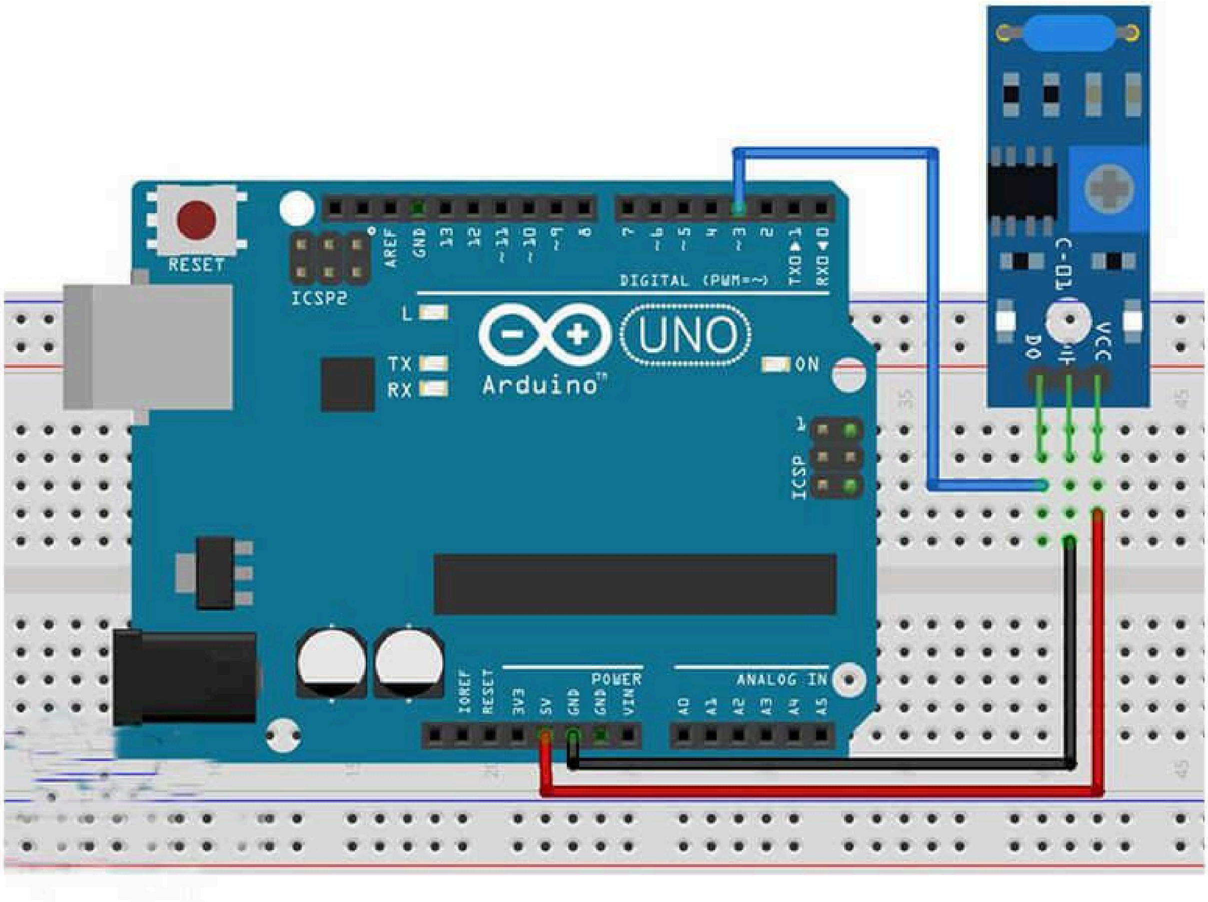
Component	Quantity	Purpose
Breadboard	1	Easier prototyping
LED	1	Extra visual alert
220Ω Resistor	1	LED protection
ESP32	1	Future IoT integration
Power Bank/Battery	1	Portable setup

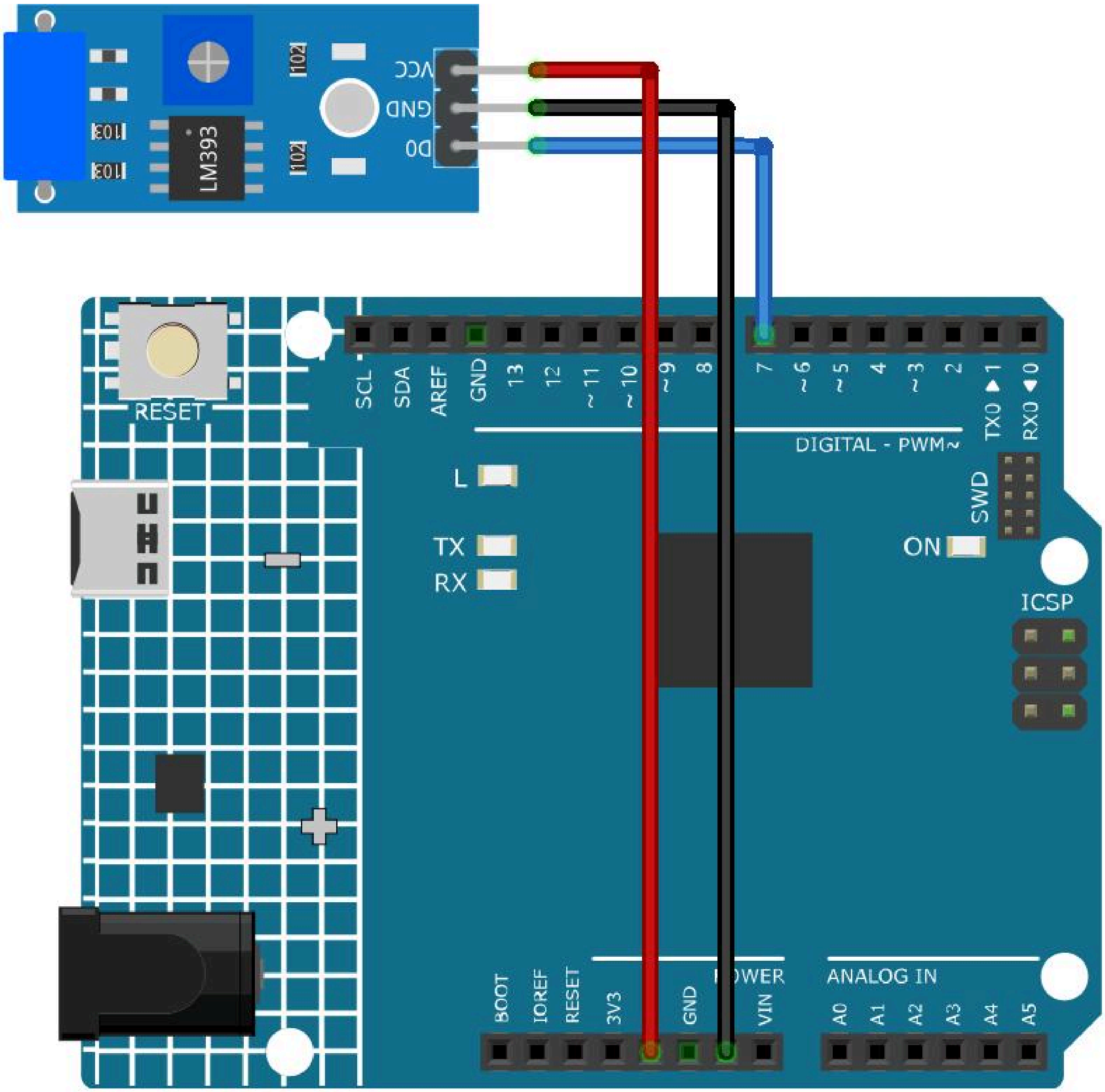
📳 4. Component Explanation

📳 SW-420 Vibration Sensor

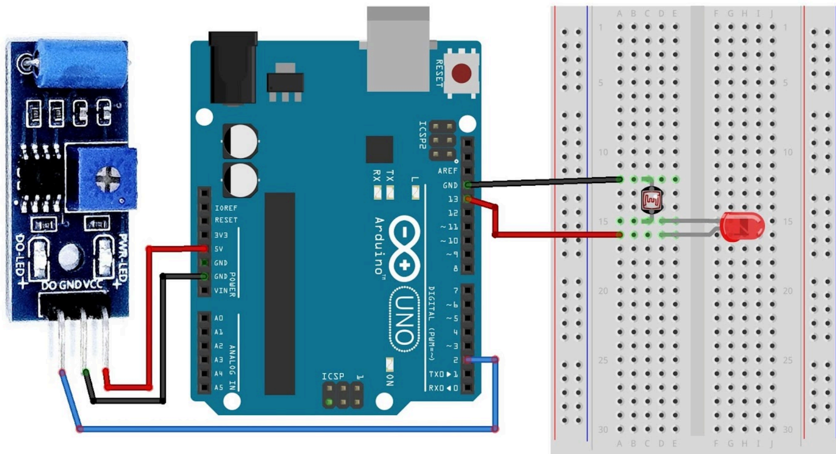

P103
U1
VCC
DO


Vcc (3.3V to 5V)
GND
Digital Output
VCC
DO
C-01

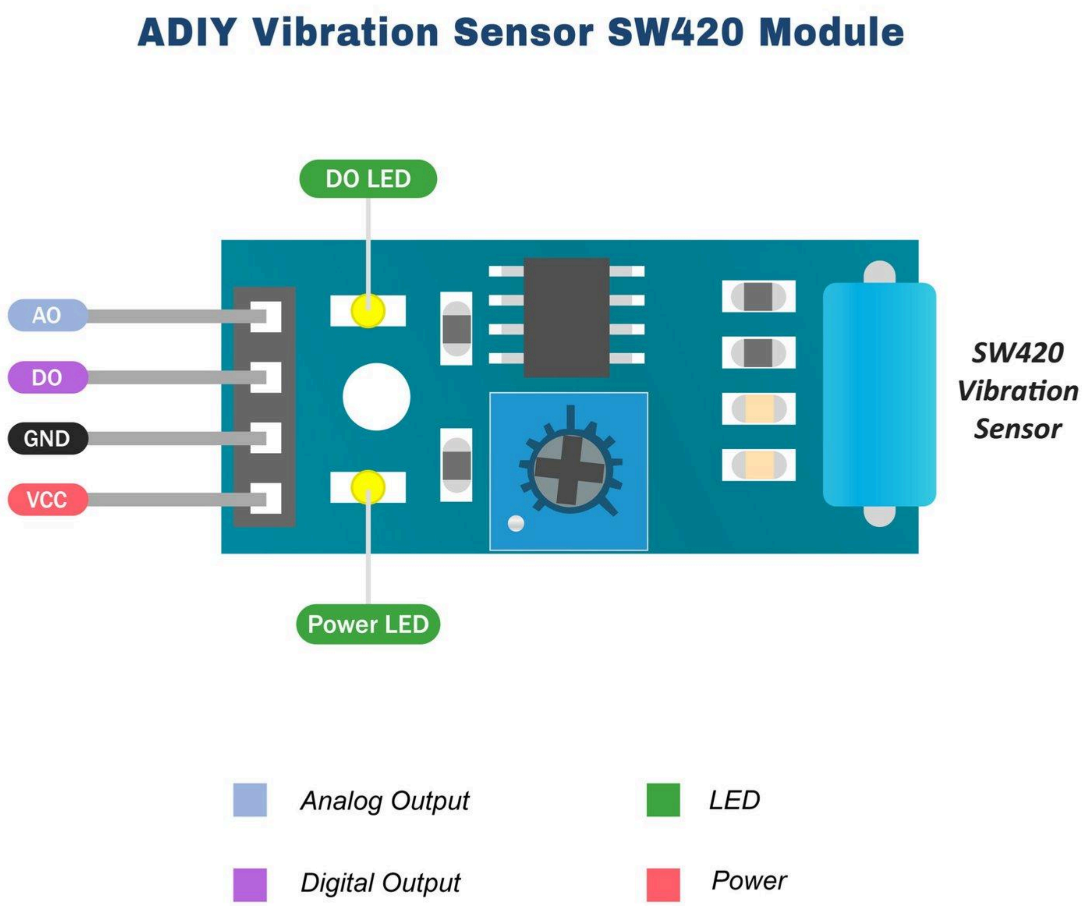

RESET
ICSP2
AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
4
~3
2
TX0▶1
RX0◀0
DIGITAL (PWM=~)
L
TX
RX
UNO
Arduino
ON
ICSP
IOREF
RESET
3V3
5V
GND
GND
VIN
POWER
ANALOG IN
A0
A1
A2
A3
A4
A5
D0
VCC



VCC
GND
DO
LM393
RESET
SCL
SDA
AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
4
~3
2
TX0 ▶1
RX0 ◀0
DIGITAL - PWM~
L
TX
RX
ON
SWD
ICSP
BOOT
IOREF
RESET
3V3
GND
VIN
ANALOG IN
A0
A1
A2
A3
A4
A5



DO GND VCC
PWR-LED
DO-LED
RESET
ICSP2
AREF
GND
IOREF
3V3
5V
VIN
POWER
ANALOG IN
A0
A1
A2
A3
A4
A5
TX
RX
Arduino
UNO
DIGITAL (PWM~)
ICSP
ON
TX0
RX0
A B C D E
F G H I J



Purpose

Detects:

- Shaking
- Knocking
- Physical tampering

Working Principle

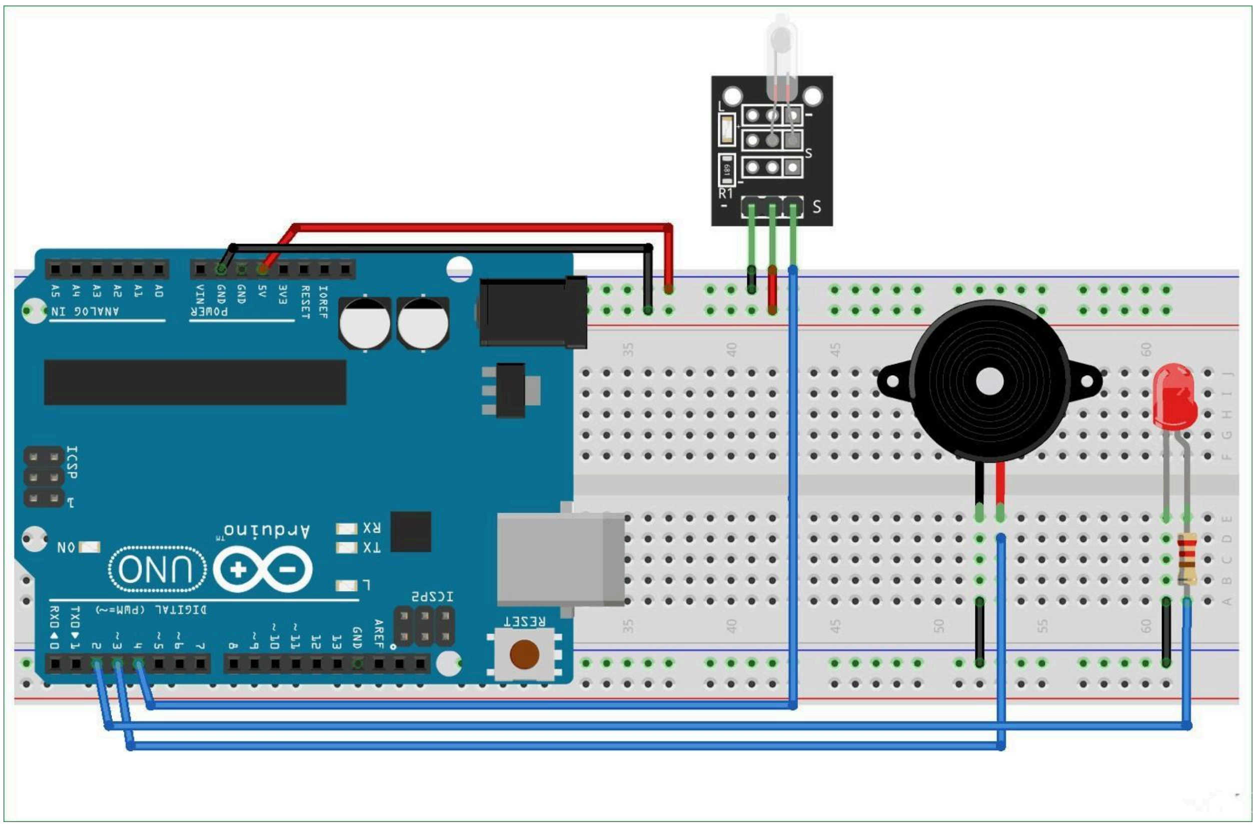
Inside the module:

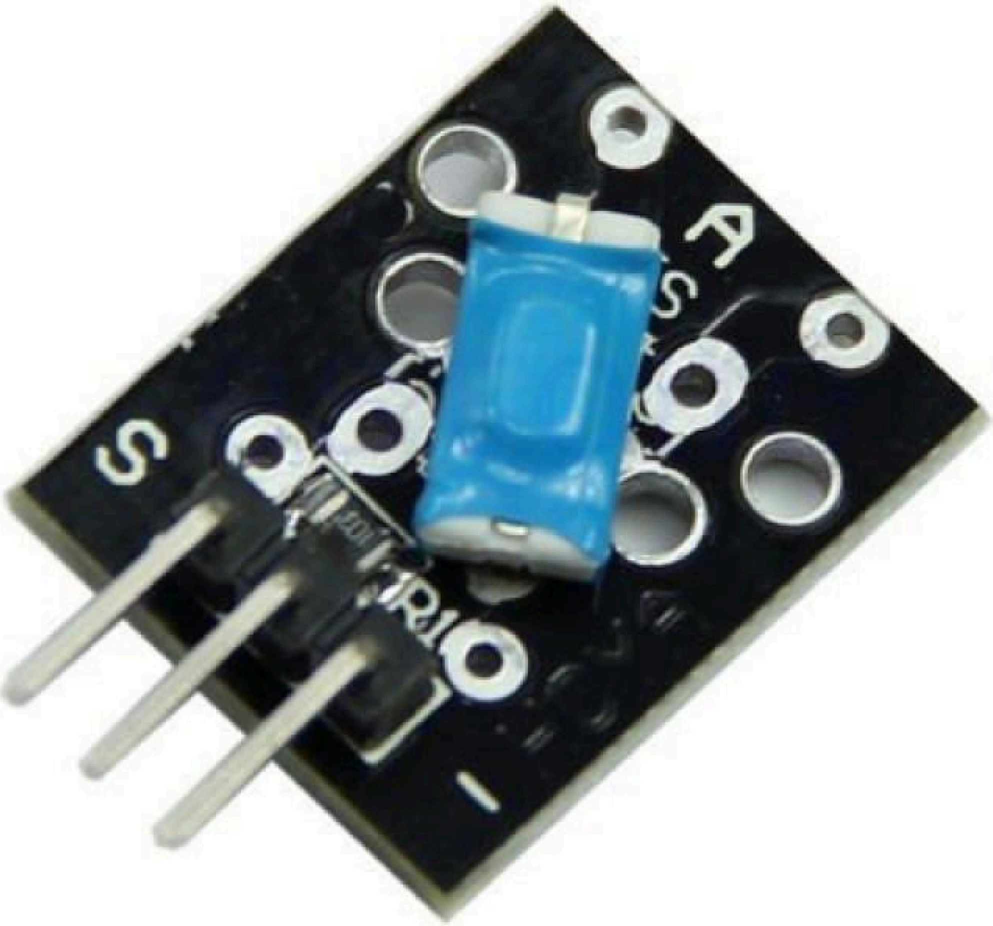
- Spring + metal contact

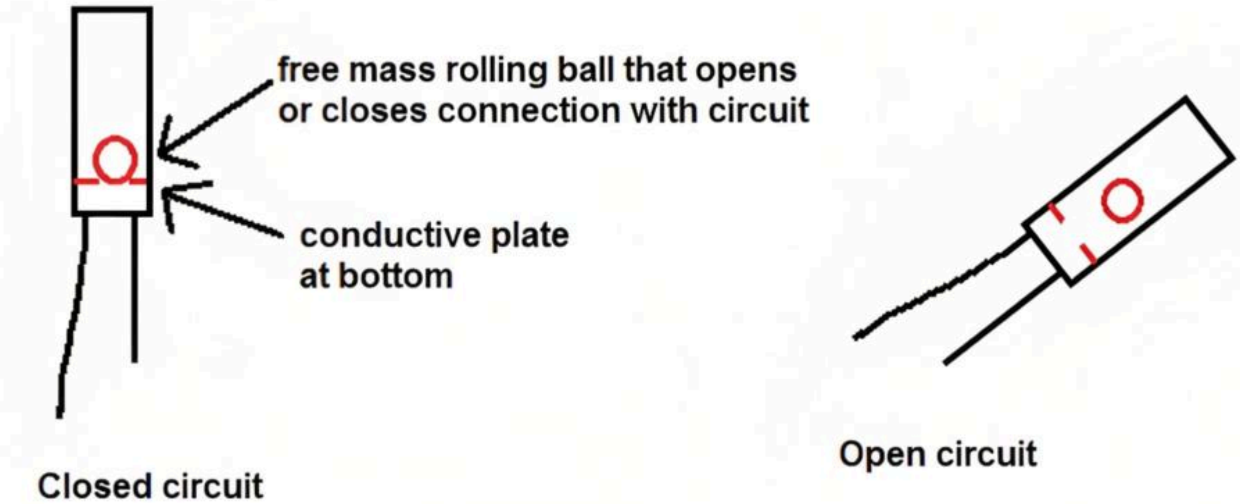
When vibration occurs:

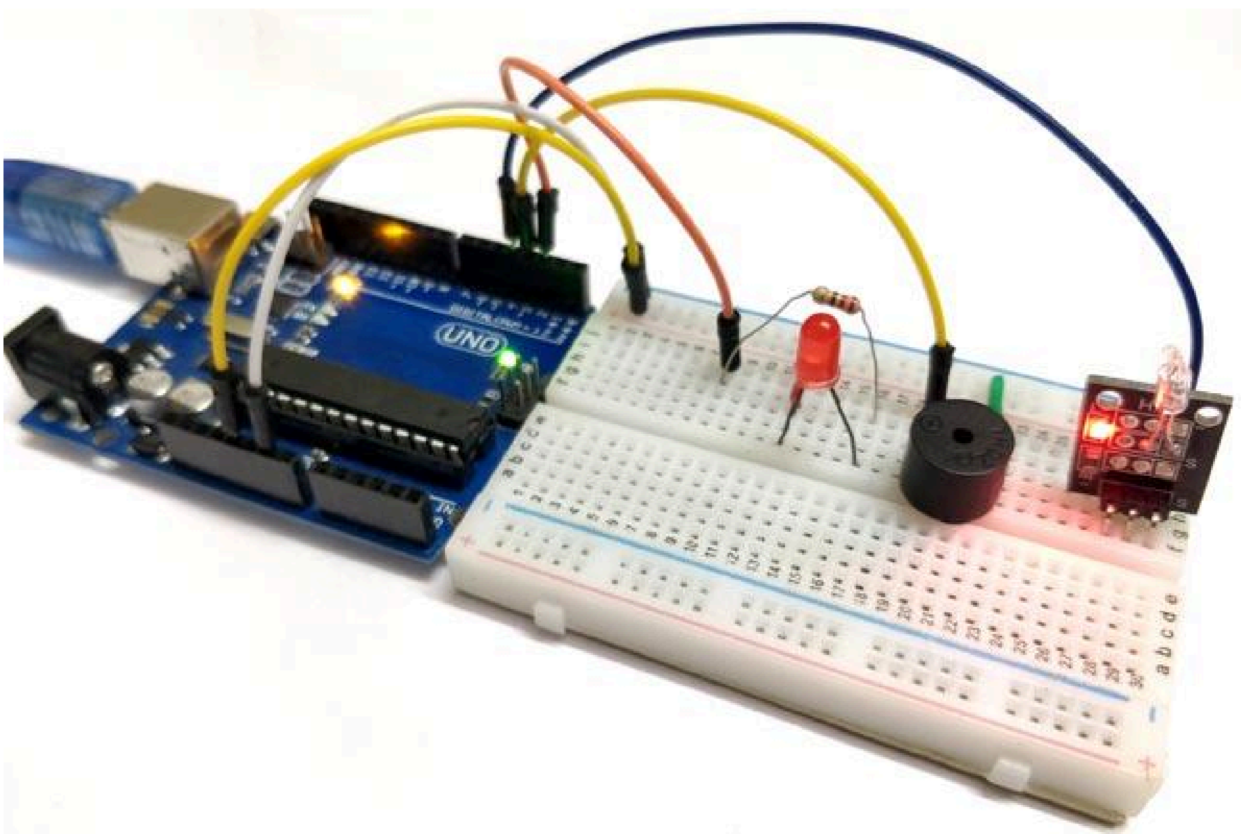
- Contact changes
- Output signal becomes HIGH

Tilt Sensor

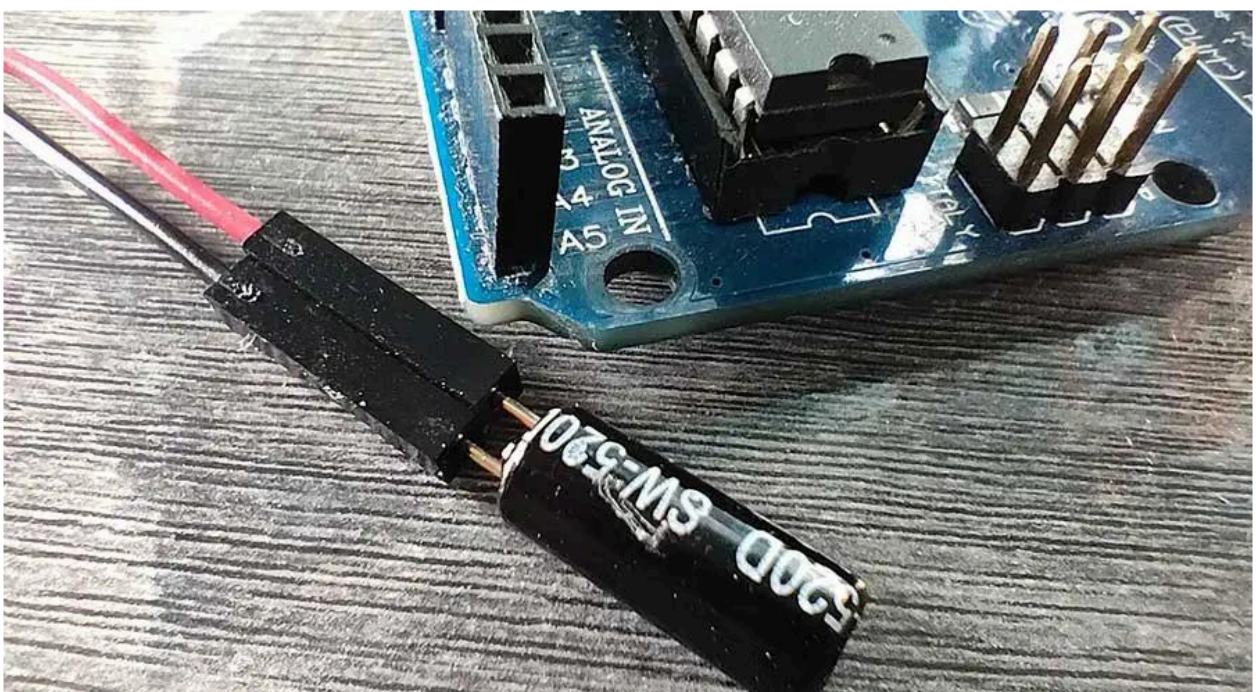


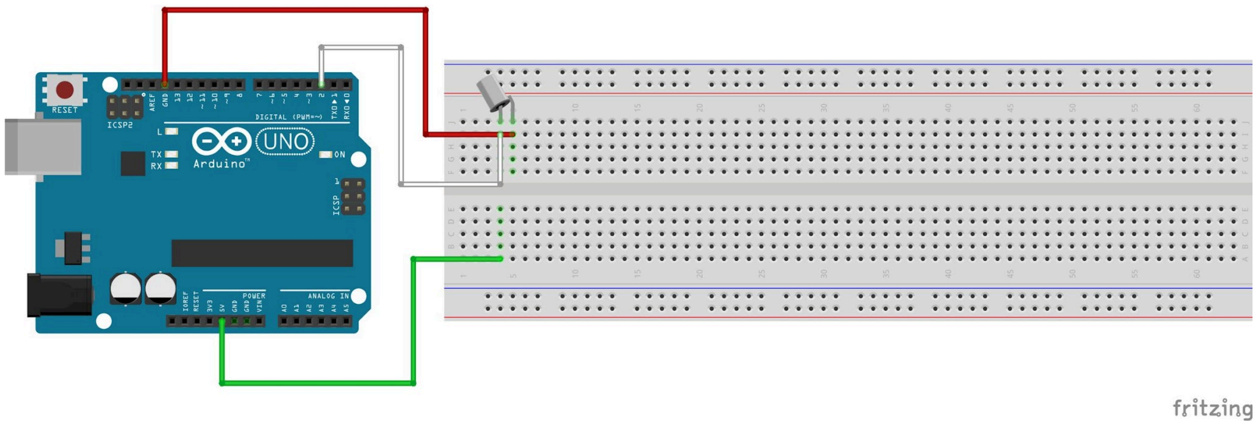

S


free mass rolling ball that opens
or closes connection with circuit
conductive plate
at bottom
Closed circuit
Open circuit


UNO


Digital
Tilt Sensor


ANALOG IN
A4
A5
SW-520D



Purpose

Detects:

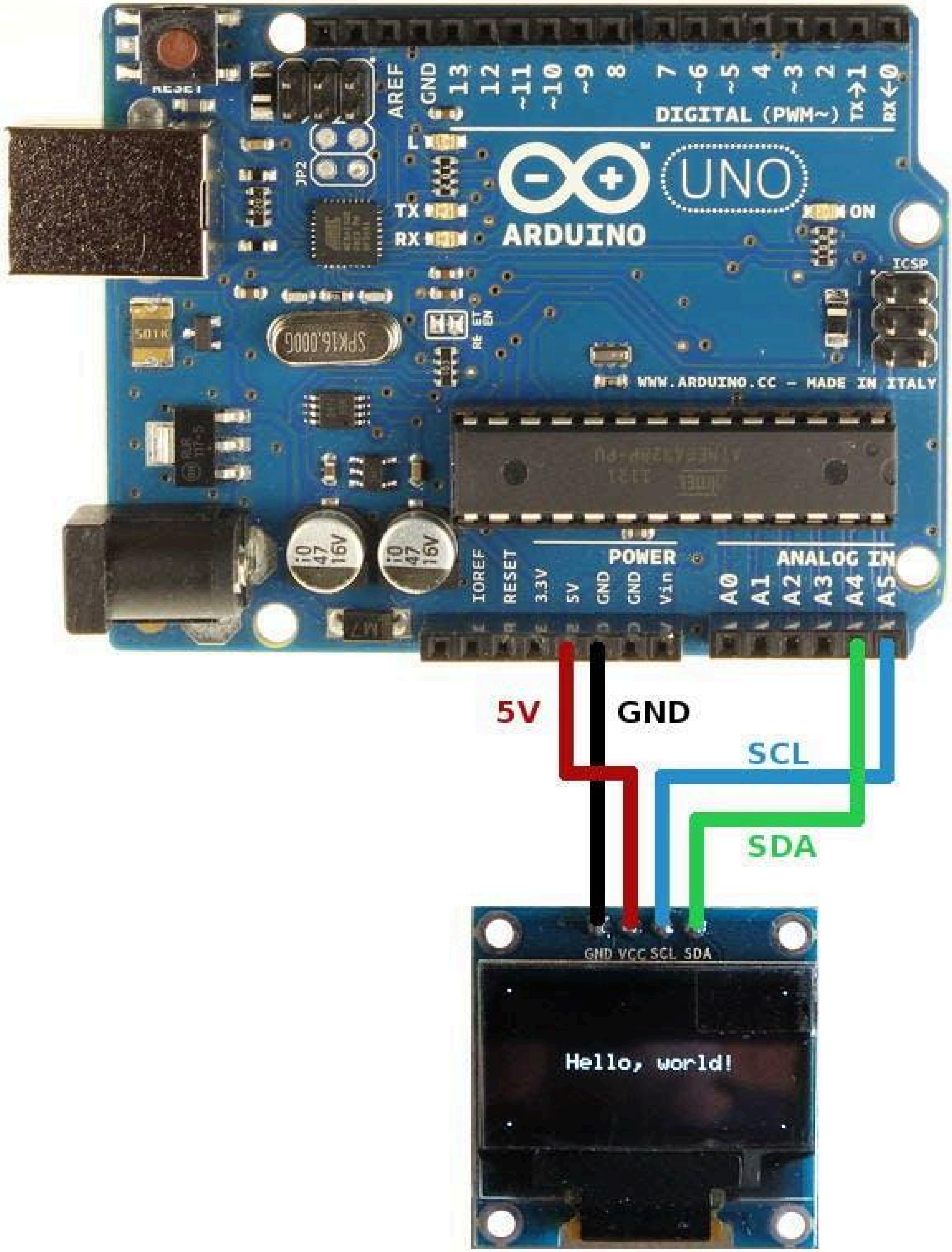
- Vehicle lifting
- Towing
- Angle/orientation change

Working Principle

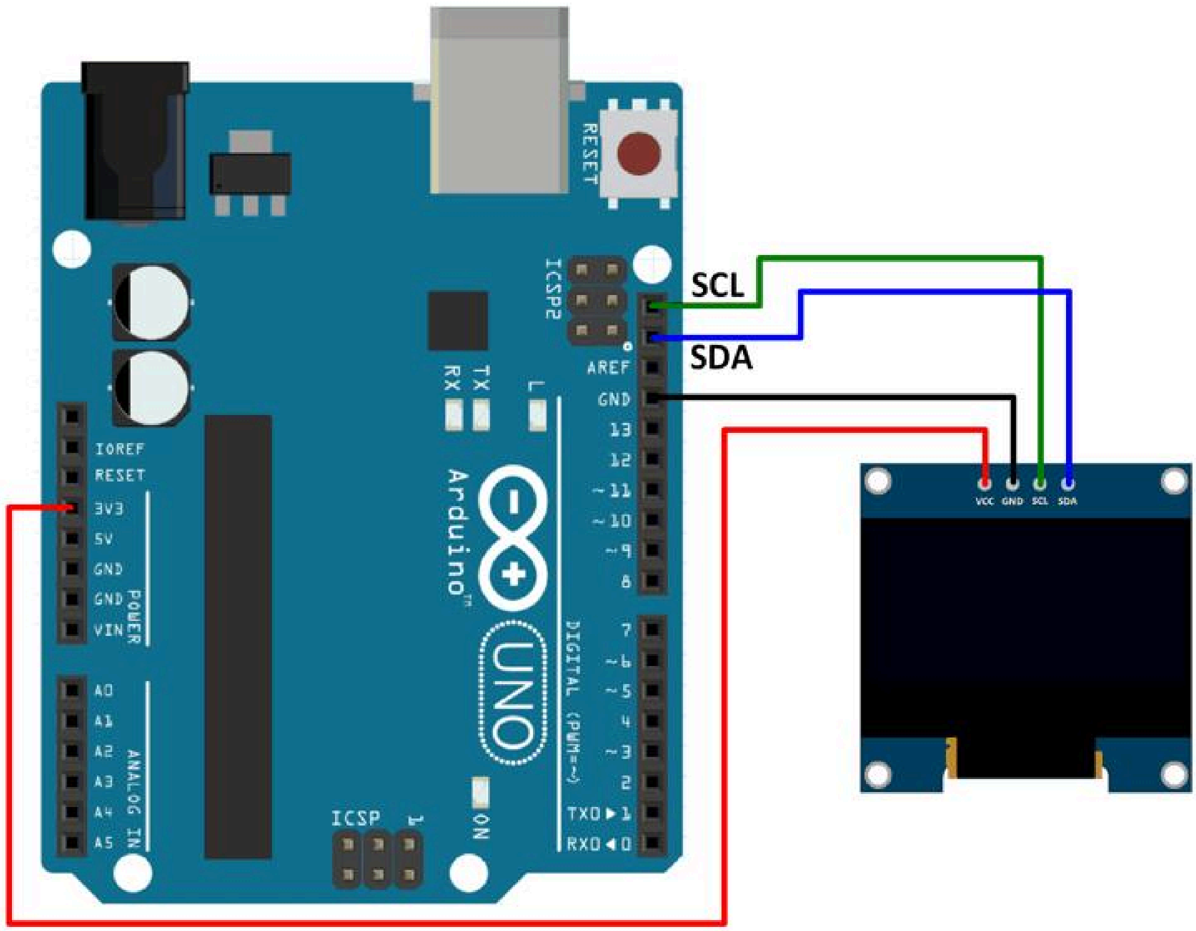
Inside the sensor:

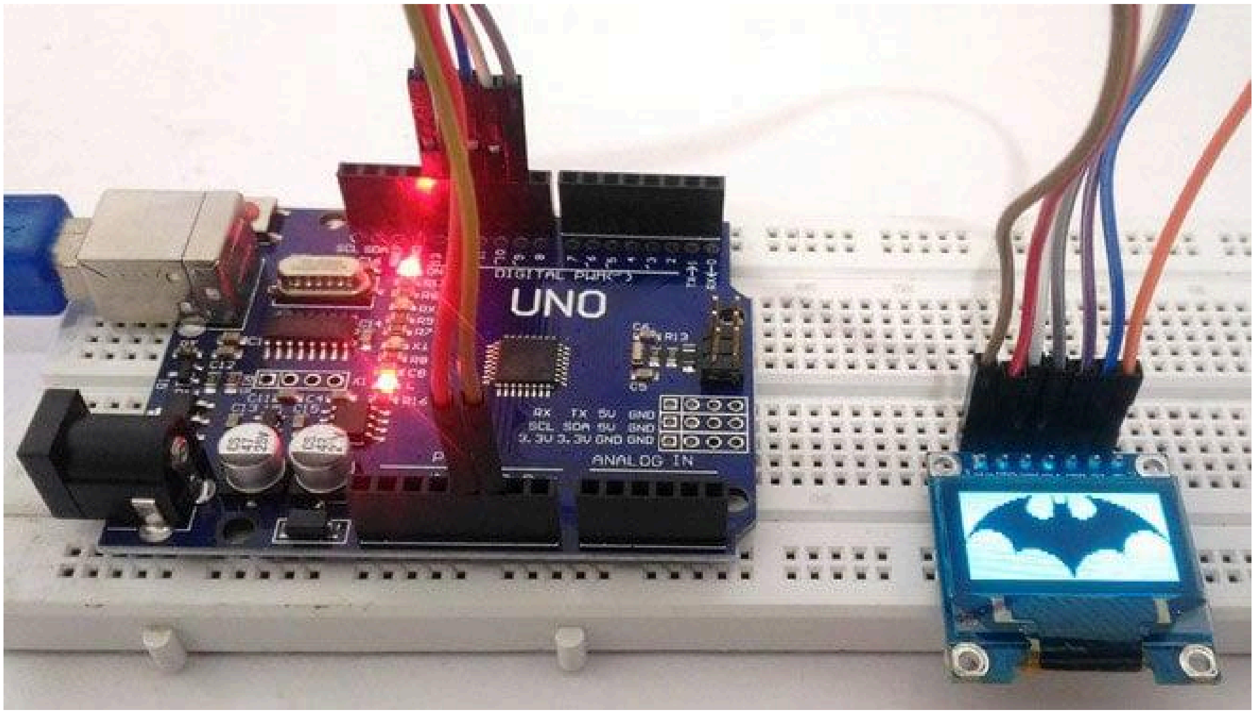
- Metal ball moves with angle change
- Circuit opens/closes

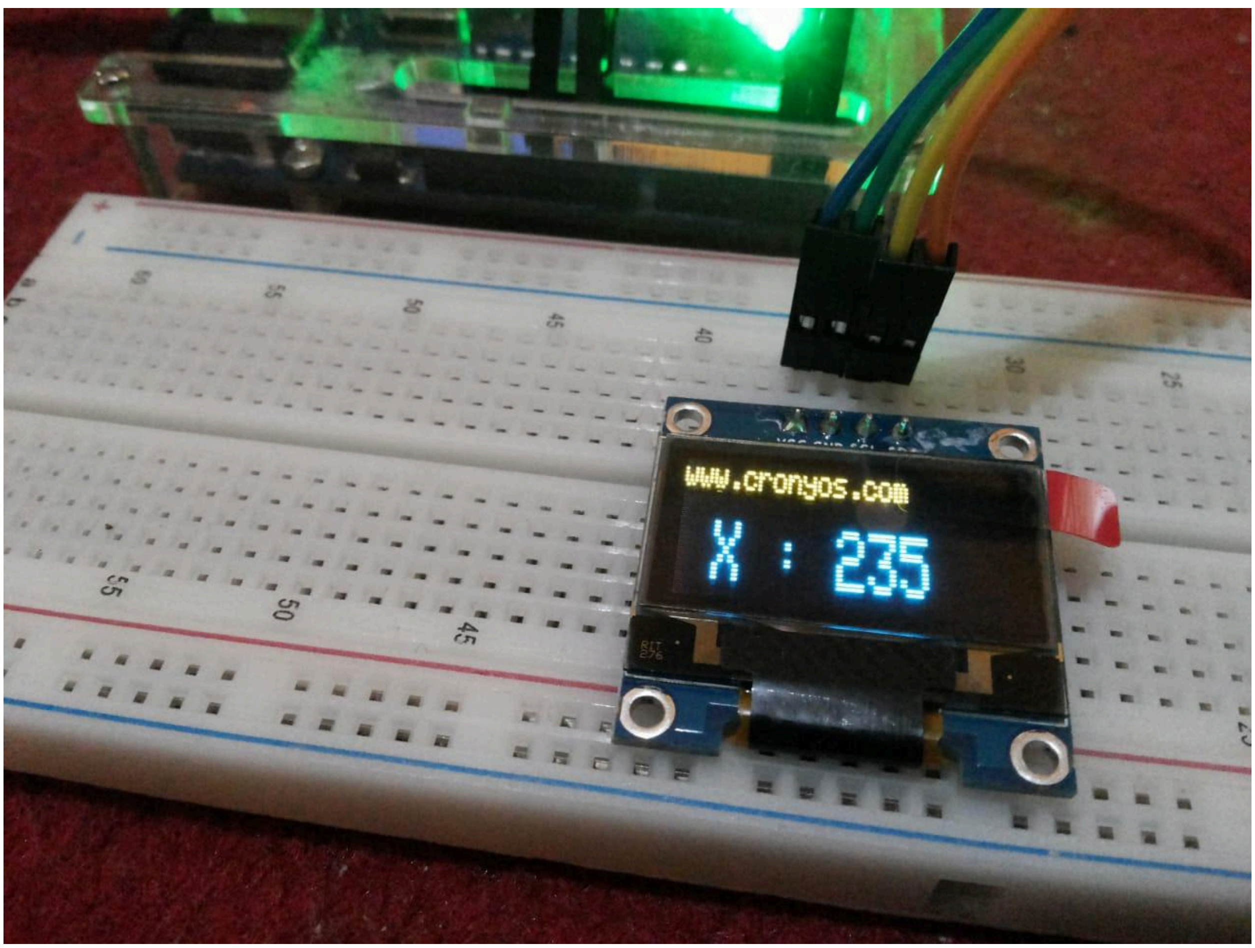
📺 OLED Display (SSD1306)


5V
GND
SCL
SDA
GND VCC SCL SDA
Hello, world!


GND VCC SCL SDA


SCL
SDA
VCC
GND
SCL
SDA
Arduino
UNO


UNO
ANALOG IN

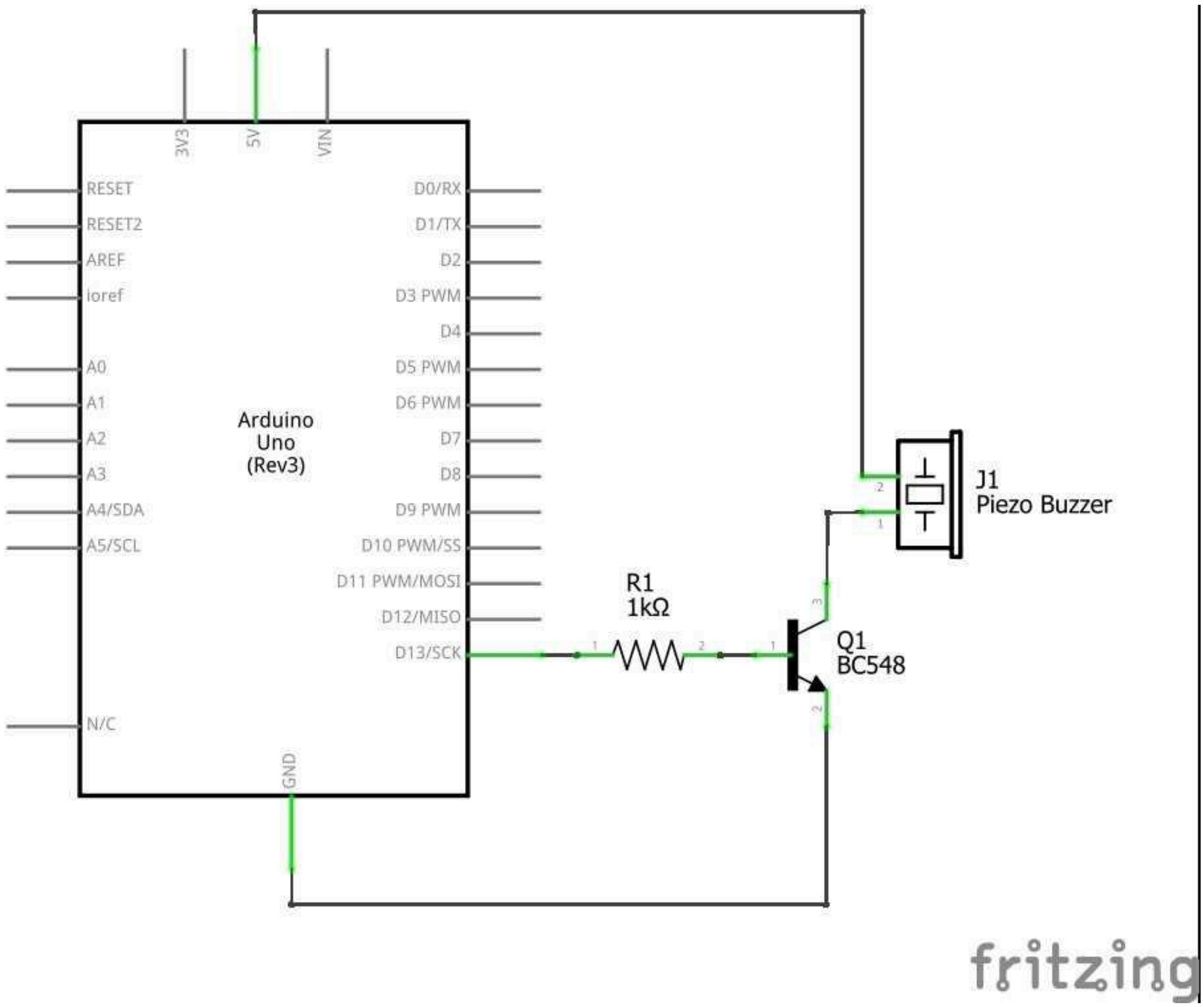

www.cronyos.com
X : 235

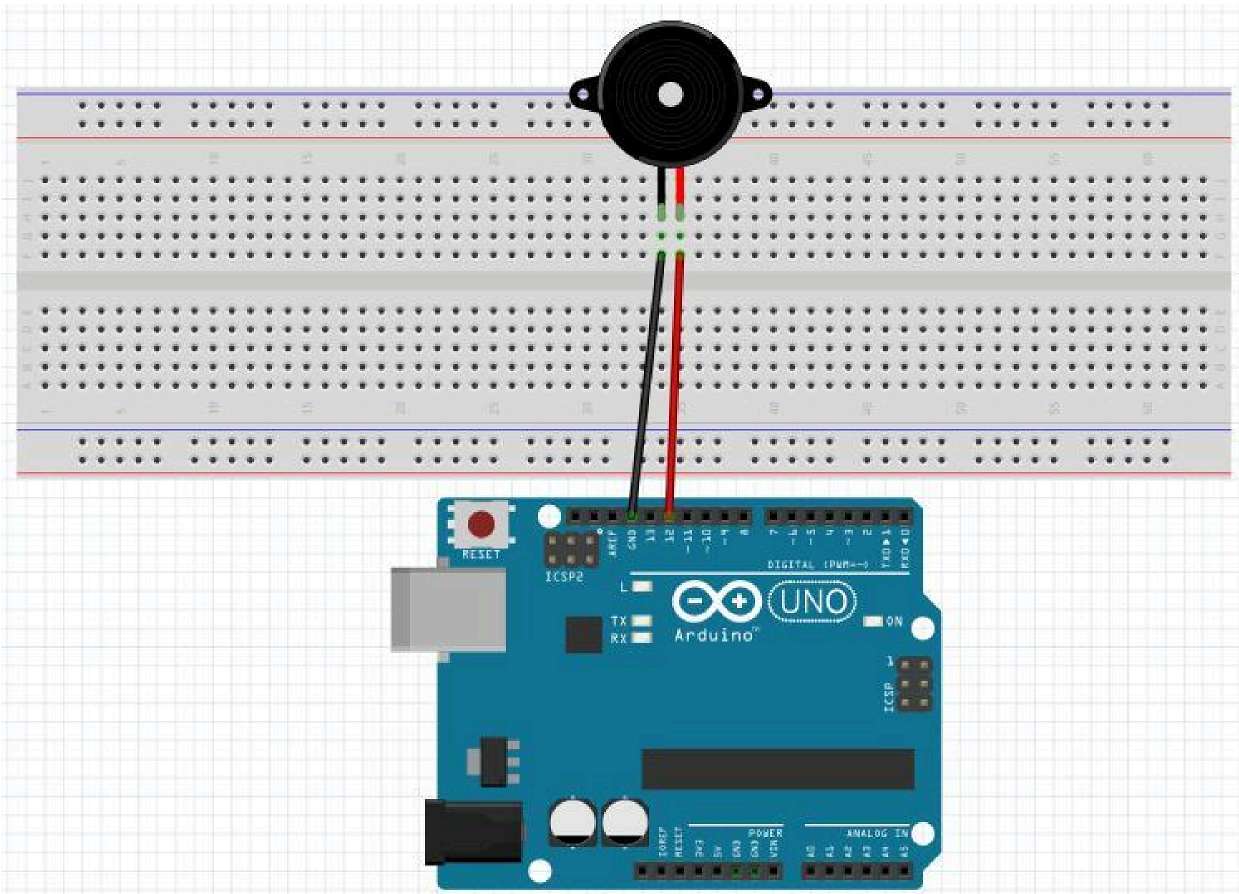
Purpose

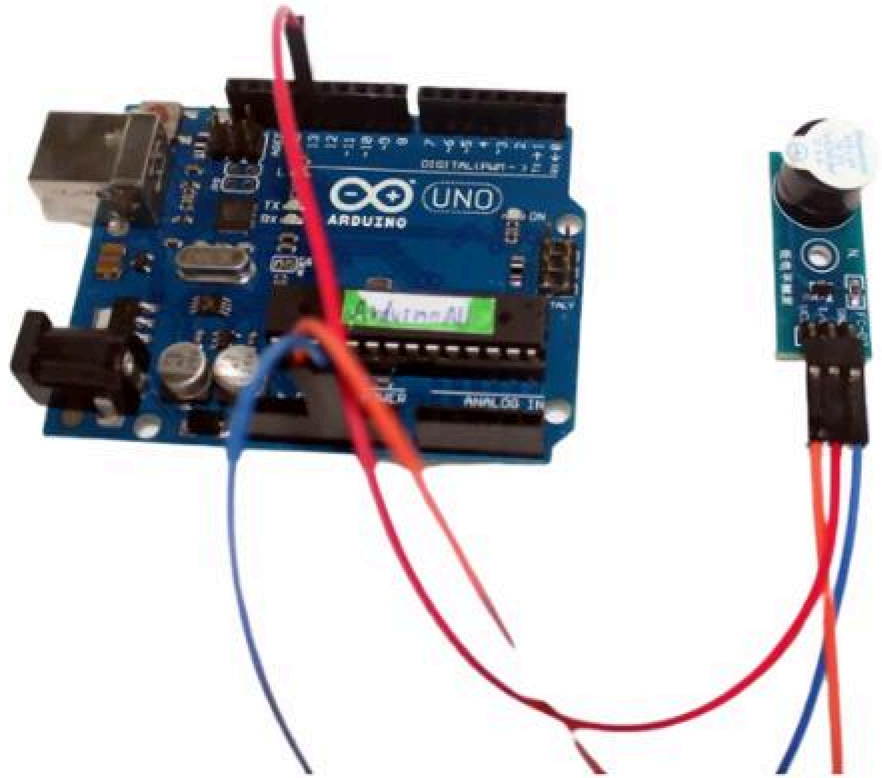
Displays:

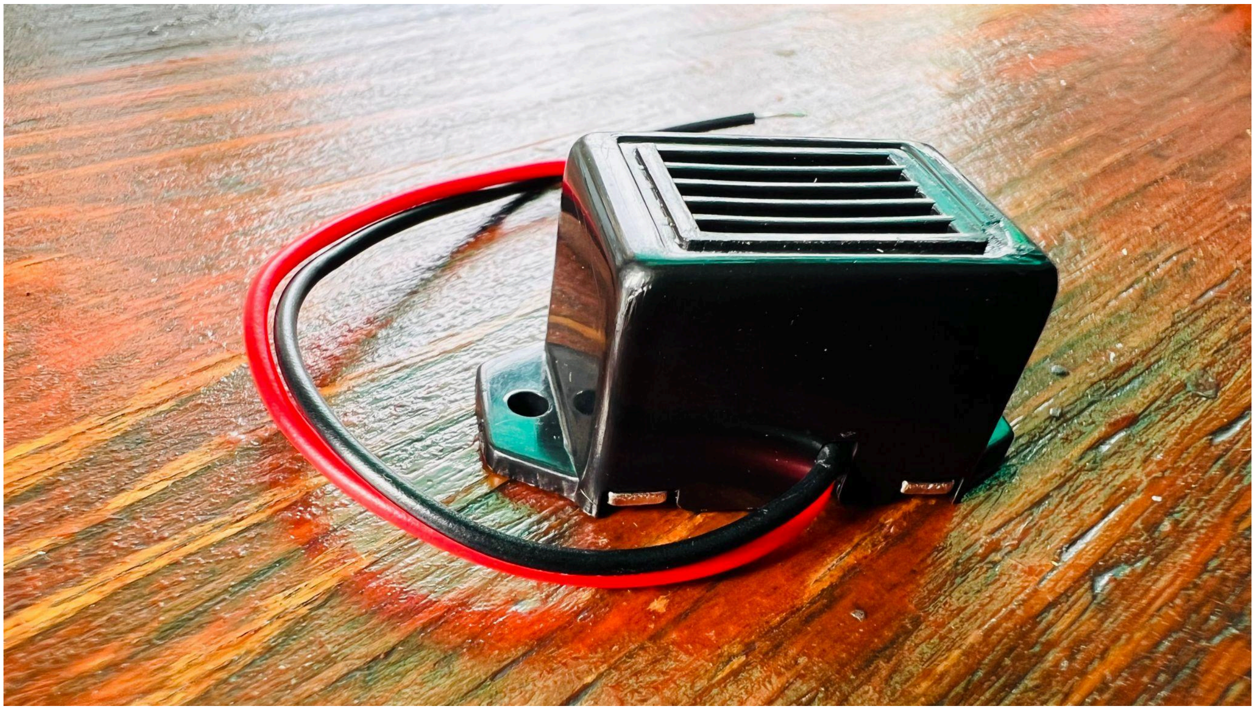
- Vehicle Safe
- ALERT! Movement Detected

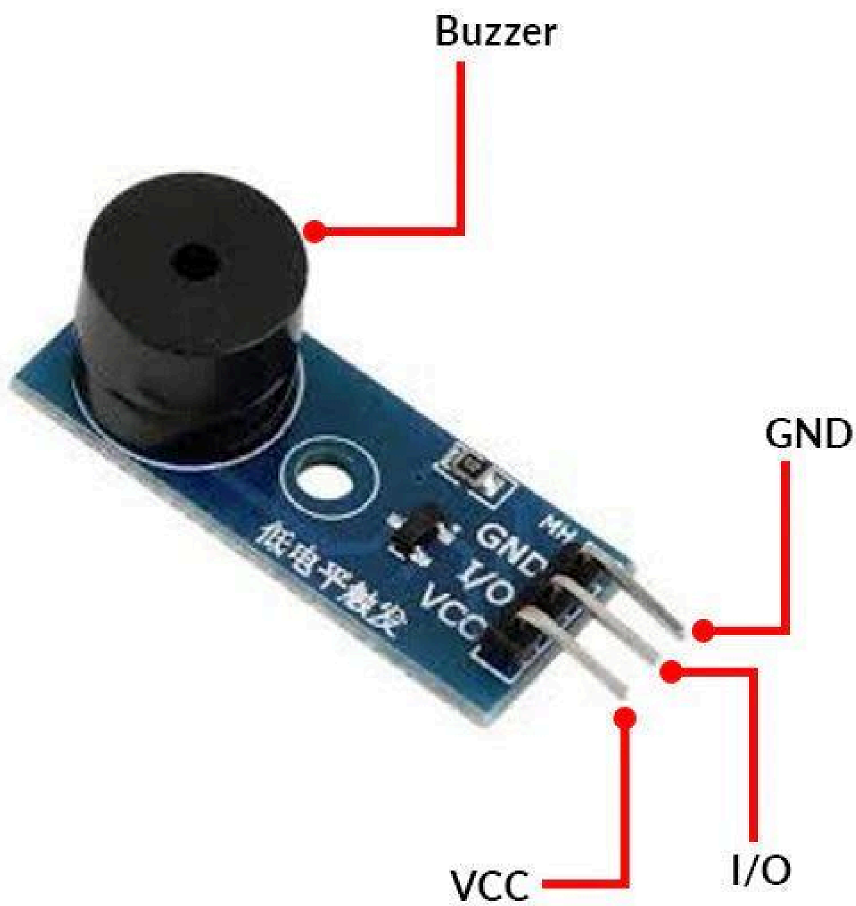
Active Buzzer



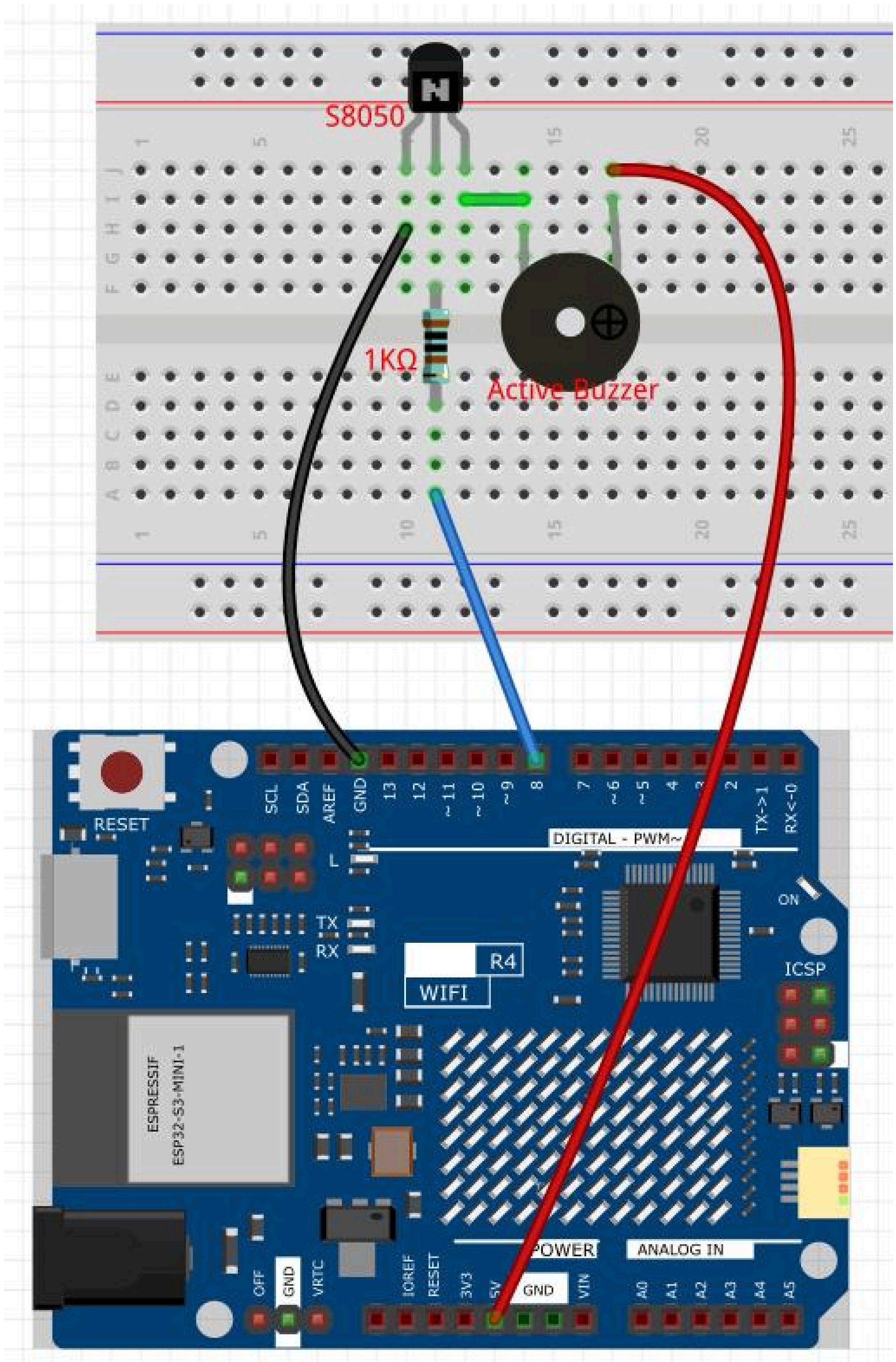

RESET
ICSP2
AREF
GND
13
12
DIGITAL (PWM~)
L
TX
RX
UNO
Arduino
ON
ICSP
IOREF
RESET
3V3
5V
GND
GND
VIN
POWER
ANALOG IN
A0
A1
A2
A3
A4
A5


UNO
ARDUINO
ANALOG IN





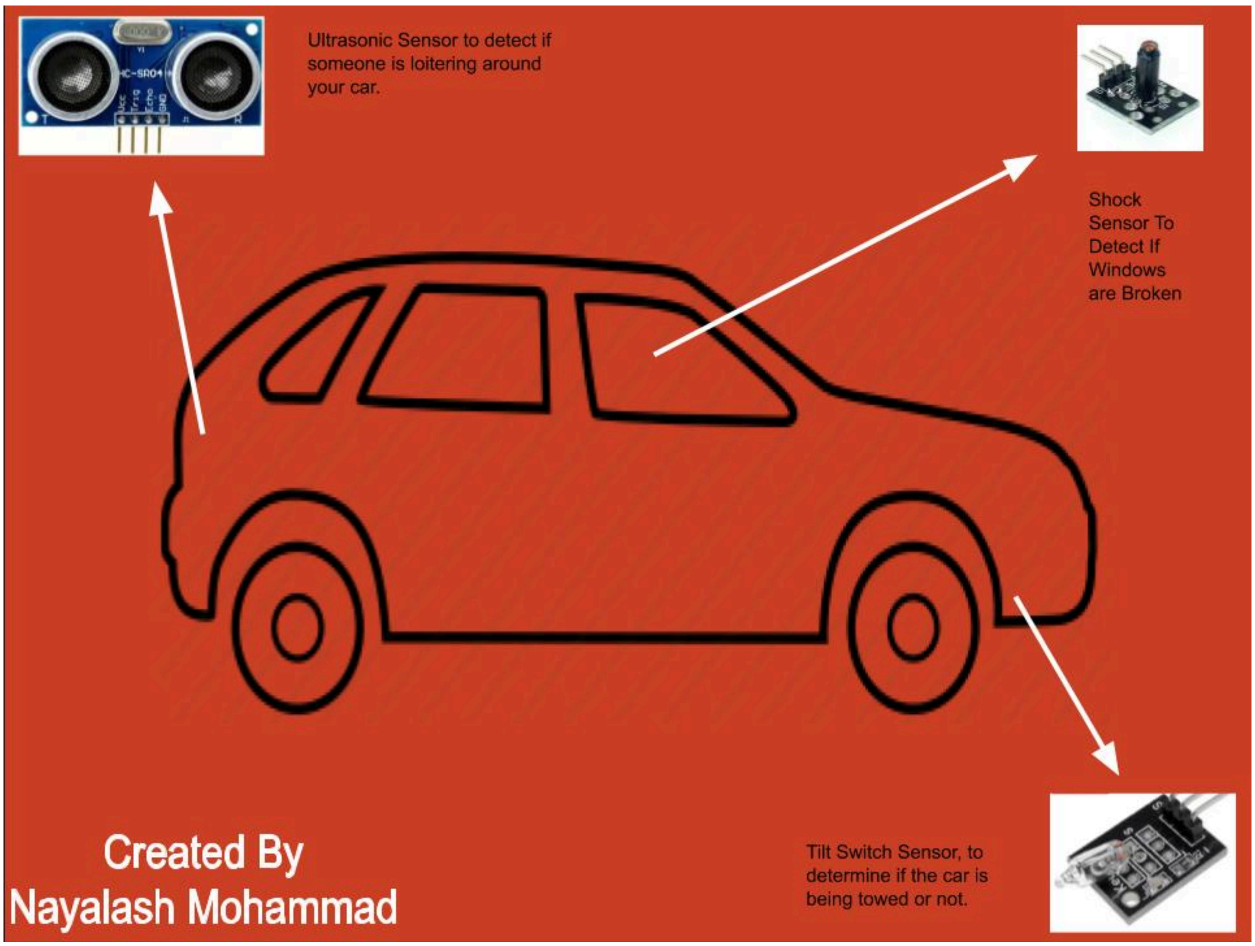
Buzzer
GND
I/O
VCC
低电平触发
GND
I/O
VCC

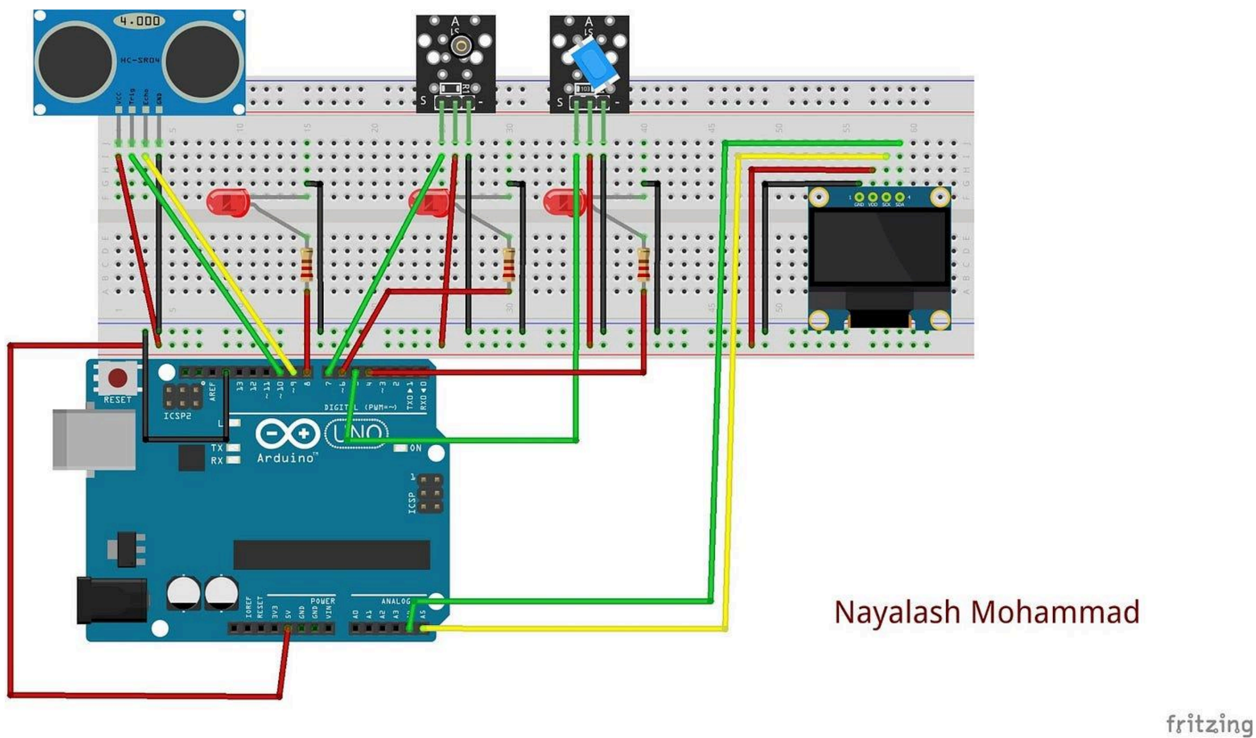

S8050
1KΩ
Active Buzzer
RESET
SCL
SDA
AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
4
~3
2
TX->1
RX<-0
DIGITAL - PWM~
ON
TX
RX
L
R4
WIFI
ICSP
ESPRESSIF
ESP32-S3-MINI-1
POWER
ANALOG IN
OFF
GND
VRTC
IOREF
RESET
3V3
5V
GND
VIN
A0
A1
A2
A3
A4
A5

Purpose

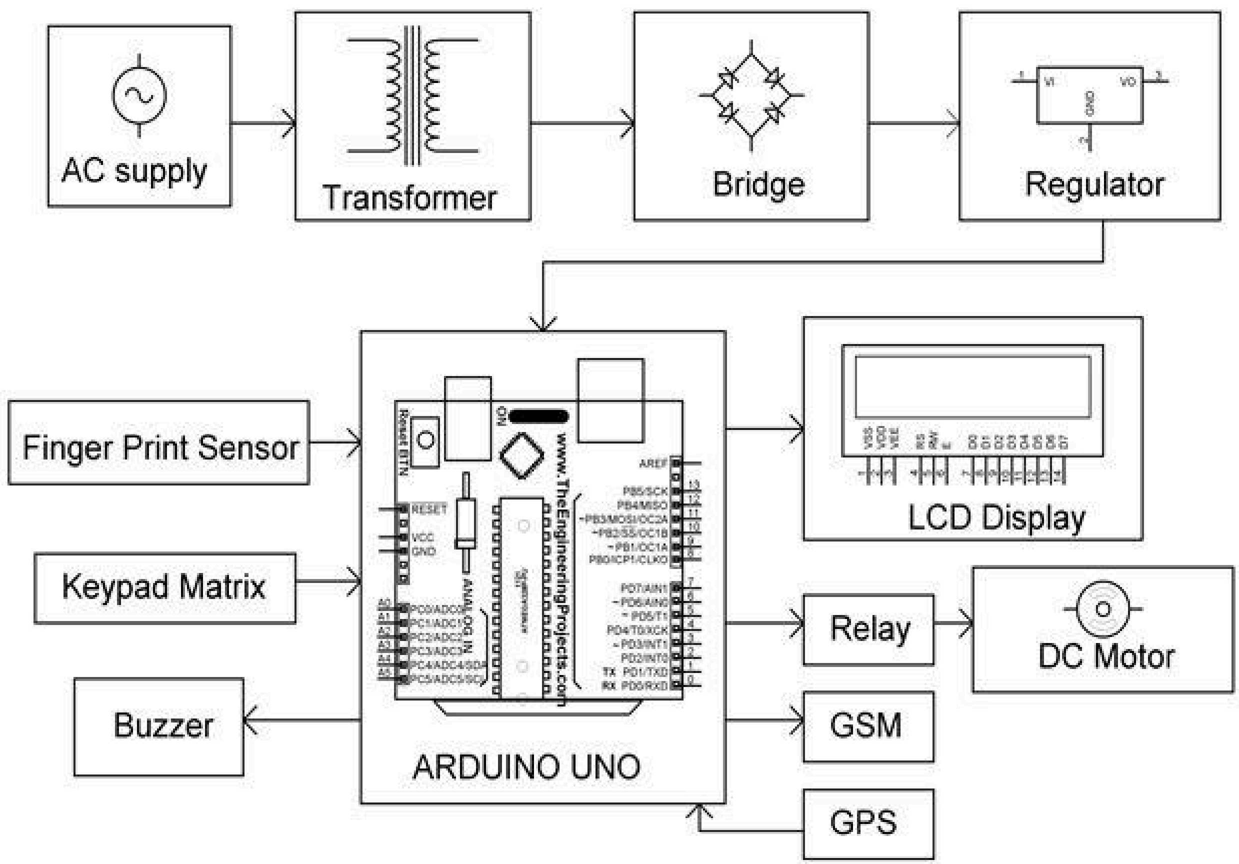
Produces alarm sound when theft is detected.

🏗️ 5. System Architecture





HC-SR04
RESET
Arduino
UNO
Nayalash Mohammad
fritzing



AC supply
Transformer
Bridge
Regulator
Finger Print Sensor
Keypad Matrix
Buzzer
www.TheEngineeringProjects.com
ARDUINO UNO
LCD Display
Relay
DC Motor
GSM
GPS

Design & Simulation

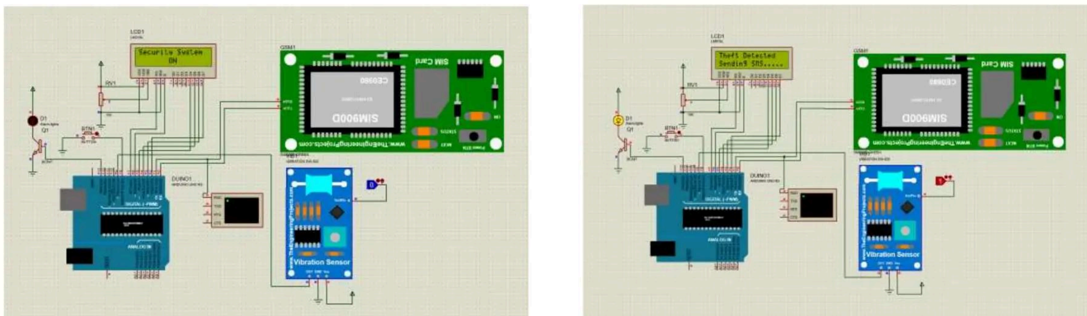


Fig3: Complete Circuit Diagram of the proposed System

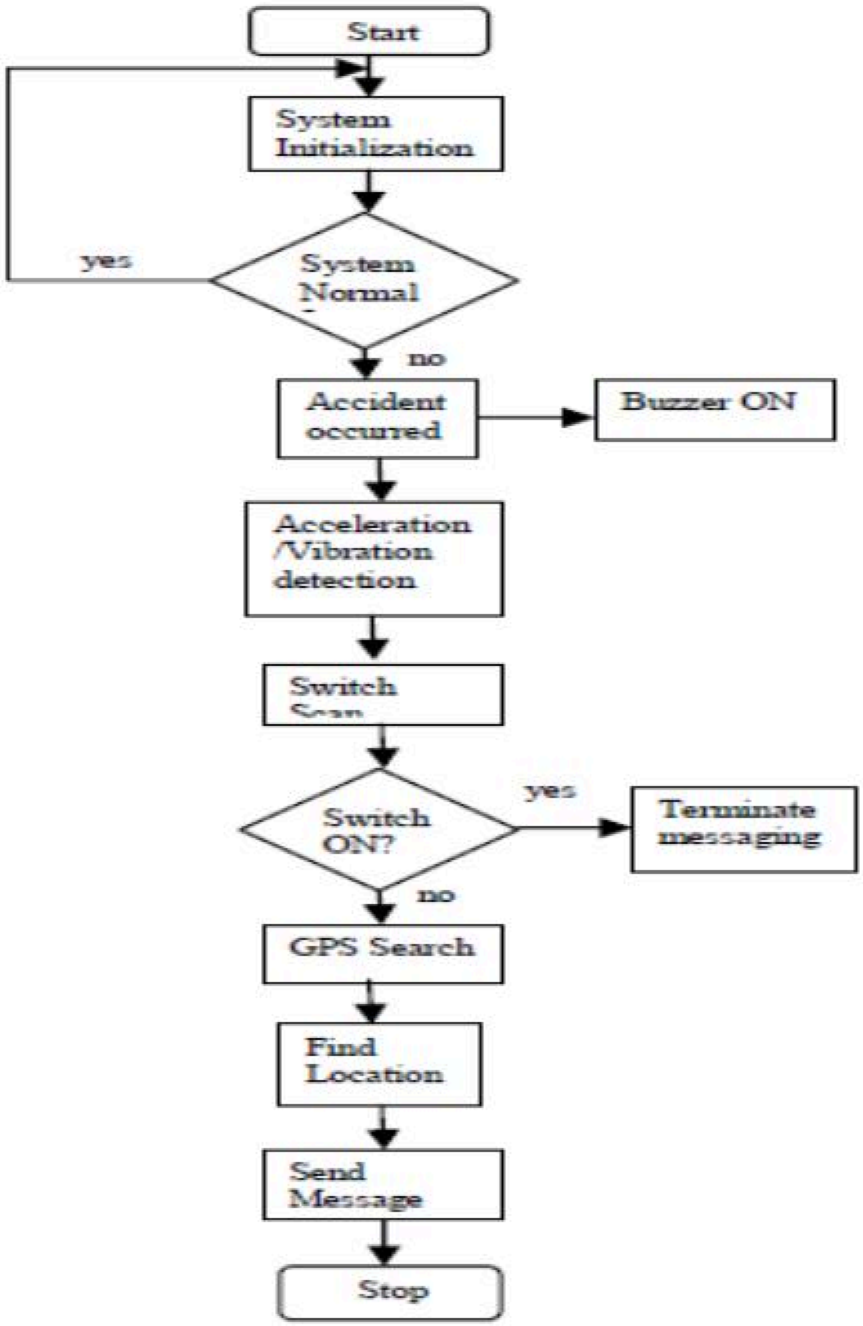
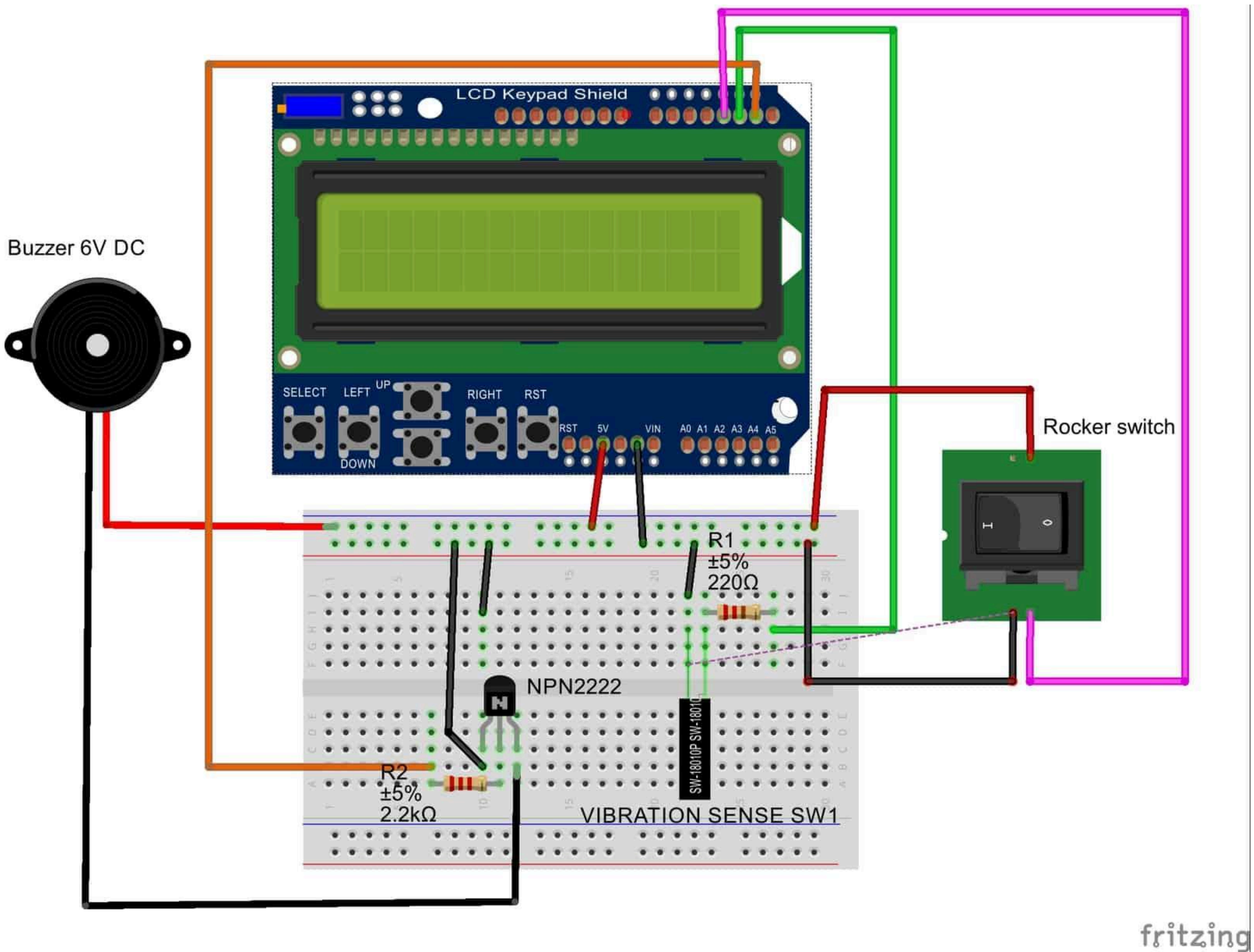
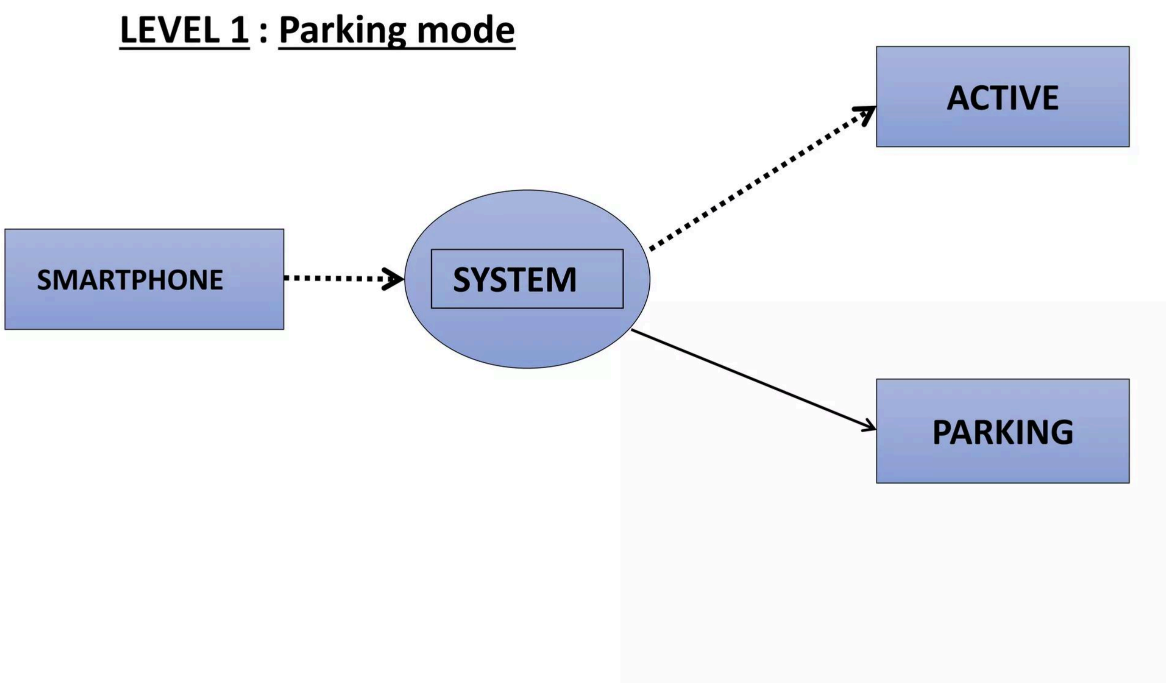


Fig. 1.1 flow chart of Vehicle monitoring system


LCD Keypad Shield
Buzzer 6V DC
SELECT
LEFT
UP
DOWN
RIGHT
RST
RST
5V
VIN
A0 A1 A2 A3 A4 A5
Rocker switch
R1
±5%
220Ω
NPN2222
R2
±5%
2.2kΩ
VIBRATION SENSE SW1
fritzing

DATA FLOW DIAGRAM



Flow

1. Sensors monitor vehicle
2. Arduino reads sensor data
3. If abnormal activity detected:
    - Trigger buzzer
    - Display alert
4. Else:
    - Show safe status

🔌 6. Circuit Connections

📺 OLED Display Connections

OLED Pin	Arduino Uno
VCC	5V
GND	GND
SDA	A4
SCL	A5

📳 Vibration Sensor Connections

SW-420 Pin	Arduino Uno
VCC	5V
GND	GND
DO	Pin 2

📐 Tilt Sensor Connections

Tilt Sensor	Arduino Uno
Pin 1	5V
Pin 2	Pin 3

👉 Configure Pin 3 as `INPUT_PULLUP`

🔊 Buzzer Connections

Buzzer	Arduino Uno

Positive (+)	Pin 8
Negative (-)	GND

💻 7. Software Required

Software	Purpose
Arduino IDE	Programming Arduino
Adafruit GFX Library	OLED graphics
Adafruit SSD1306 Library	OLED control

📥 8. Installing Libraries

In Arduino IDE:

1. Open Arduino IDE
2. Go to:
   - Sketch → Include Library → Manage Libraries
3. Search and install:
   - Adafruit GFX
   - Adafruit SSD1306

💻 9. Arduino Code

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);
```

```
#define VIBRATION_PIN 2
#define TILT_PIN 3
#define BUZZER 8

void setup() {

  pinMode(VIBRATION_PIN, INPUT);
  pinMode(TILT_PIN, INPUT_PULLUP);
  pinMode(BUZZER, OUTPUT);

  Serial.begin(9600);

  if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    Serial.println("OLED Failed");
    while(true);
  }

  display.clearDisplay();
  display.setTextSize(1);
  display.setTextColor(WHITE);

  display.setCursor(0,10);
  display.println("System Ready");
  display.display();

  delay(2000);
}

void loop() {

  int vibration = digitalRead(VIBRATION_PIN);
  int tilt = digitalRead(TILT_PIN);

  display.clearDisplay();
  display.setCursor(0,10);

  if(vibration == HIGH || tilt == LOW) {

    digitalWrite(BUZZER, HIGH);

    display.println("ALERT!");
    display.println("Movement Detected");
```

```
    Serial.println("ALERT: Theft Attempt!");

  } else {

    digitalWrite(BUZZER, LOW);

    display.println("Vehicle Safe");
  }

  display.display();

  delay(500);
}
```

🚀 10. Implementation Steps

Step 1: Assemble Components

- Connect all sensors/modules

Step 2: Install Libraries

- Install OLED libraries

Step 3: Upload Code

- Select correct board/port
- Upload code

Step 4: Power System

- Use USB cable or power bank

Step 5: Observe Output

OLED should display:

System Ready

Then:

Vehicle Safe

🧪 11. Testing Phase

✅ Test 1: Idle State

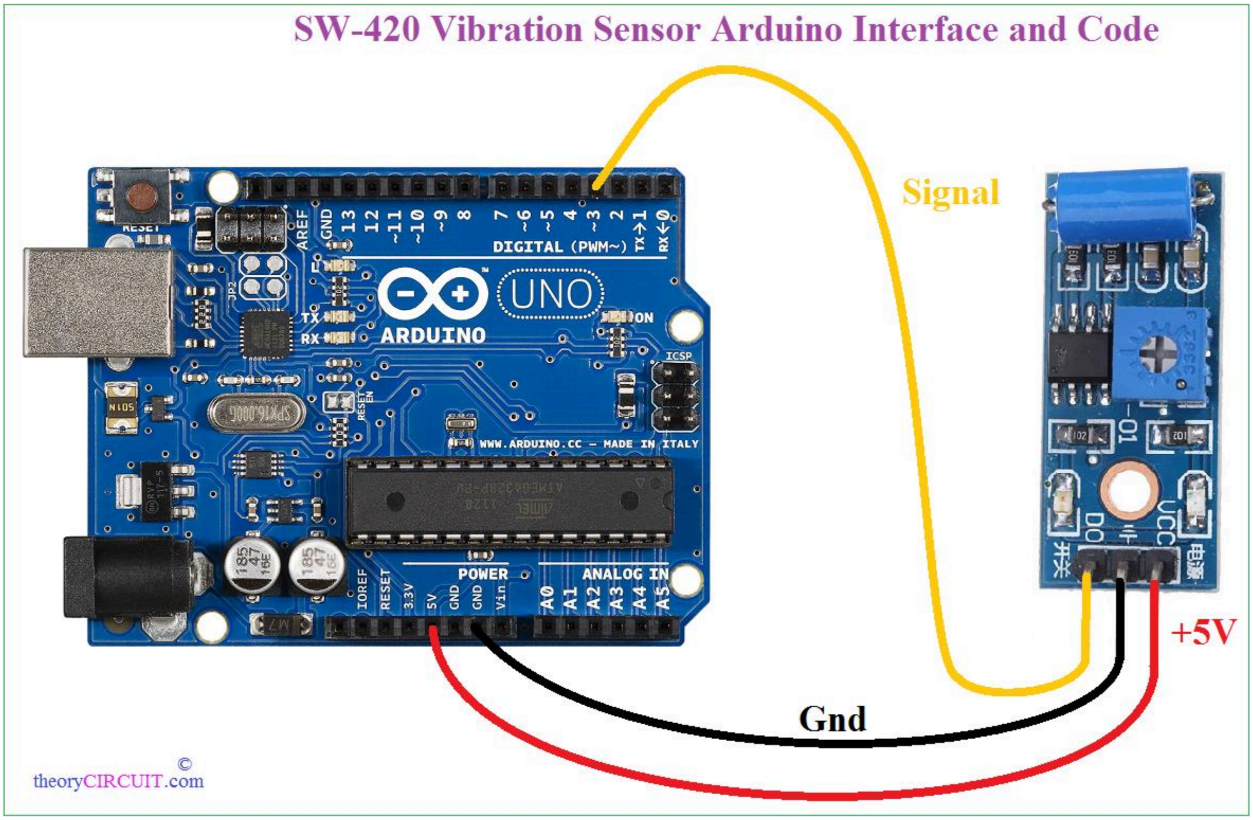
Procedure

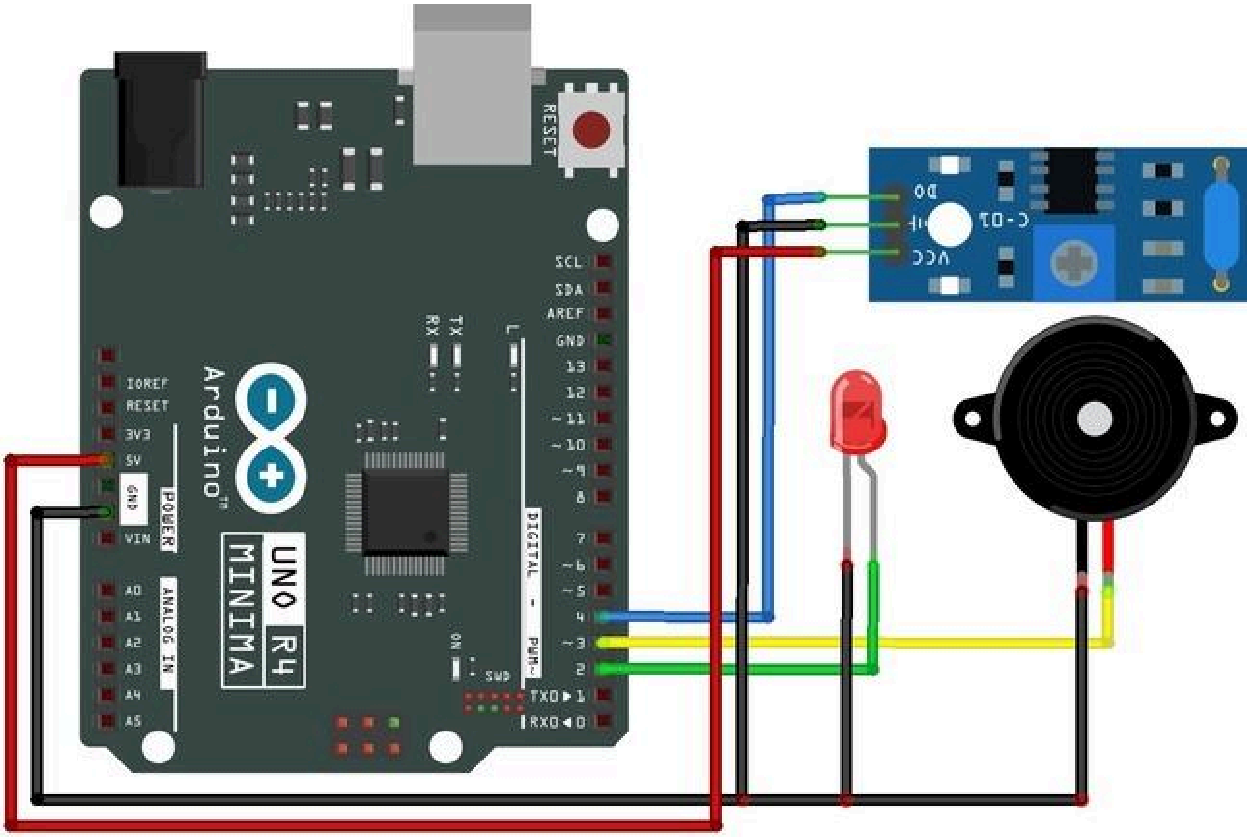
Do not touch system.

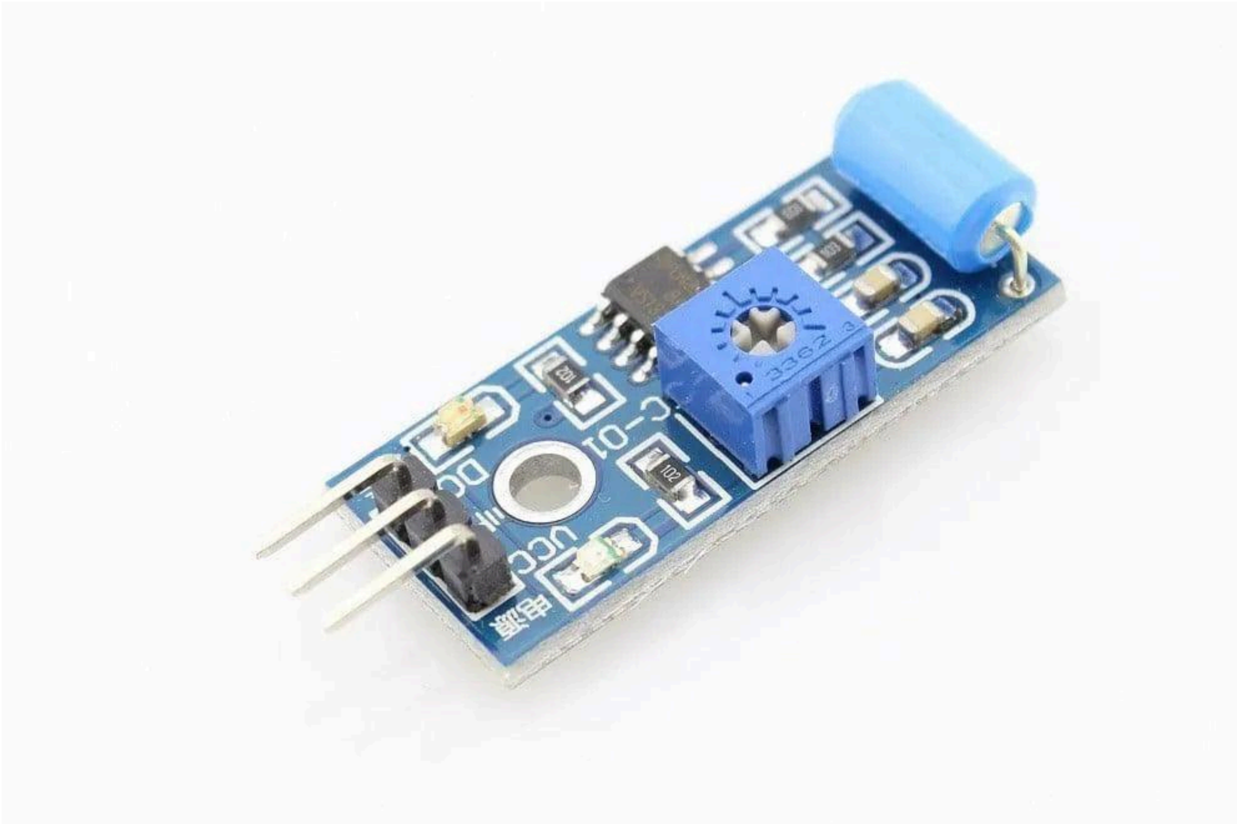
Expected Output

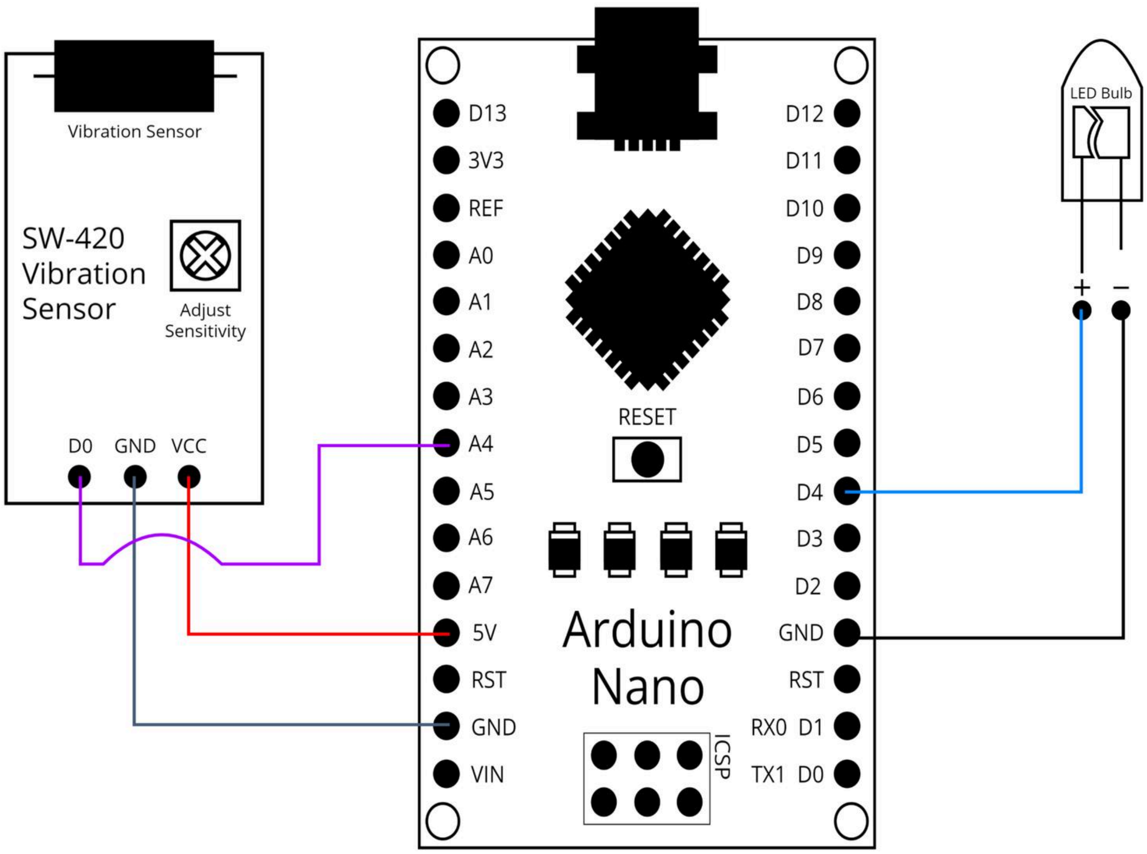
- OLED → Vehicle Safe
- Buzzer → OFF

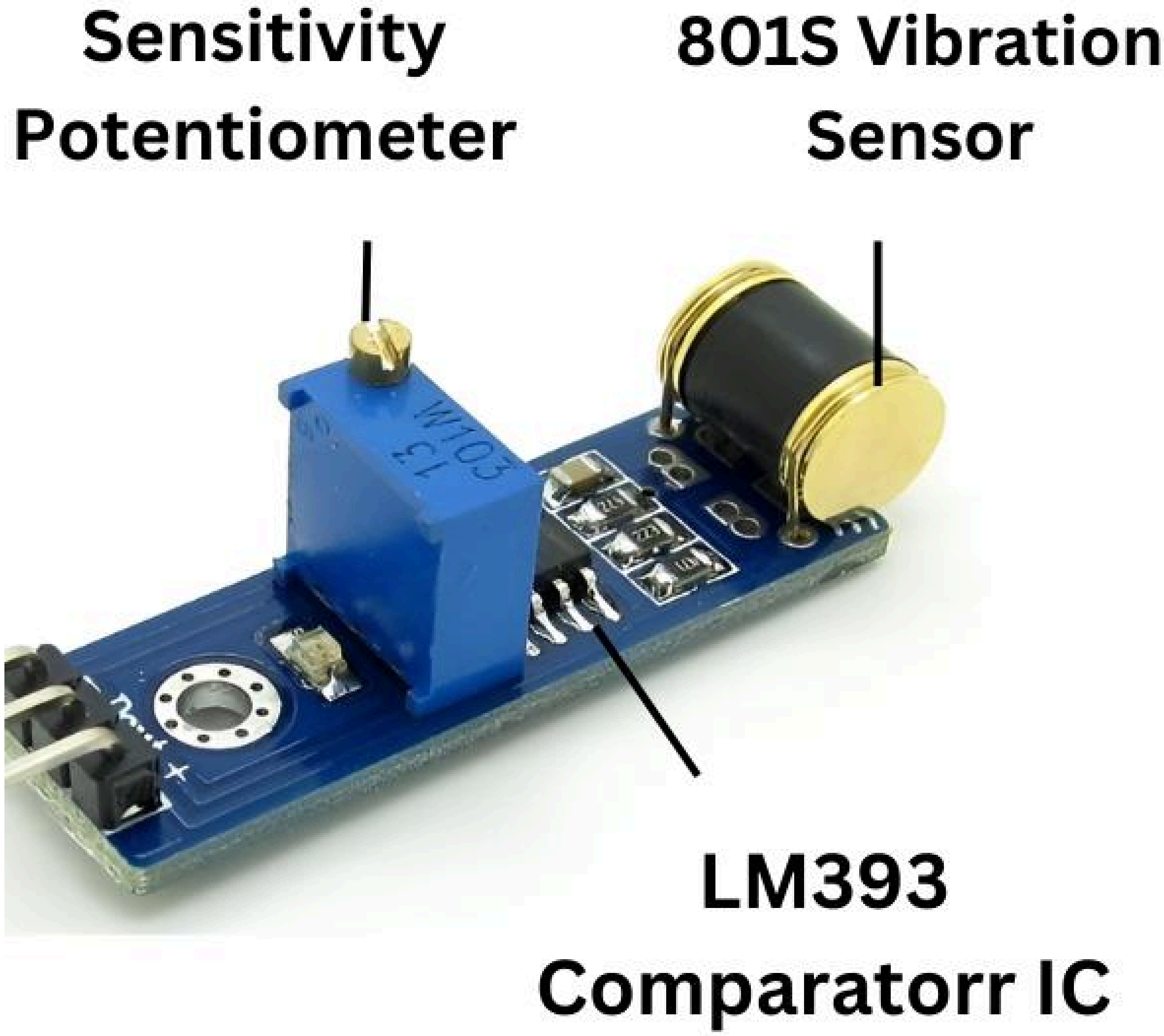
🔴 Test 2: Vibration Detection

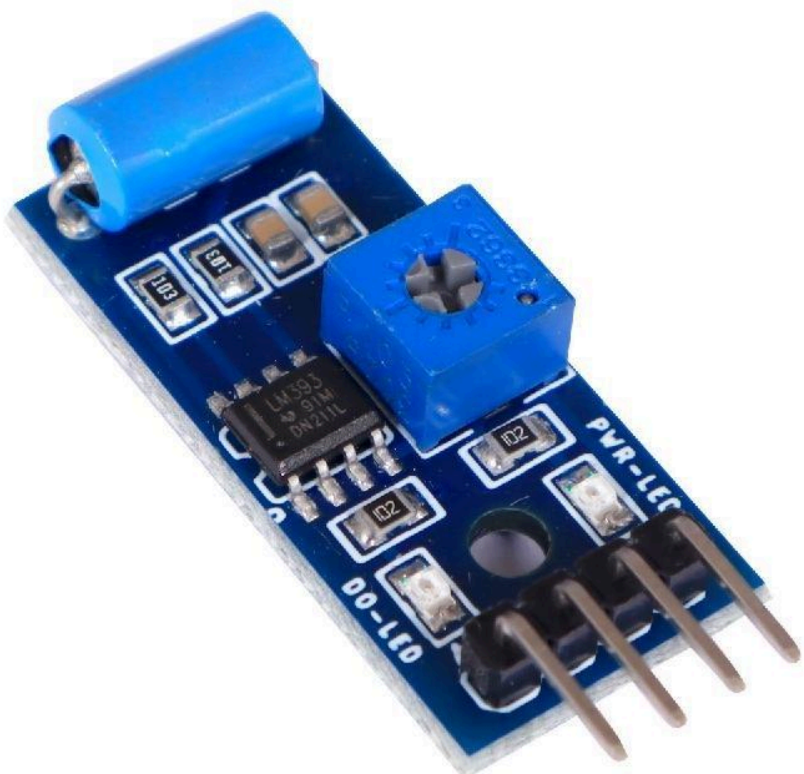

SW-420 Vibration Sensor Arduino Interface and Code
Signal
+5V
Gnd
theoryCIRCUIT.com


RESET
SCL
SDA
AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
4
~3
2
TXO▶1
RXO◀0
DIGITAL - PWM~
IOREF
RESET
3V3
5V
GND
VIN
POWER
A0
A1
A2
A3
A4
A5
ANALOG IN
Arduino
UNO R4
MINIMA
RX
TX
L
ON
SWD
DO
GND
VCC




Vibration Sensor
SW-420 Vibration Sensor
Adjust Sensitivity
D0
GND
VCC
D13
3V3
REF
A0
A1
A2
A3
A4
A5
A6
A7
5V
RST
GND
VIN
D12
D11
D10
D9
D8
D7
D6
D5
D4
D3
D2
GND
RST
RX0 D1
TX1 D0
RESET
Arduino Nano
ICSP
LED Bulb
+
−


Sensitivity Potentiometer
801S Vibration Sensor
LM393 Comparatorr IC



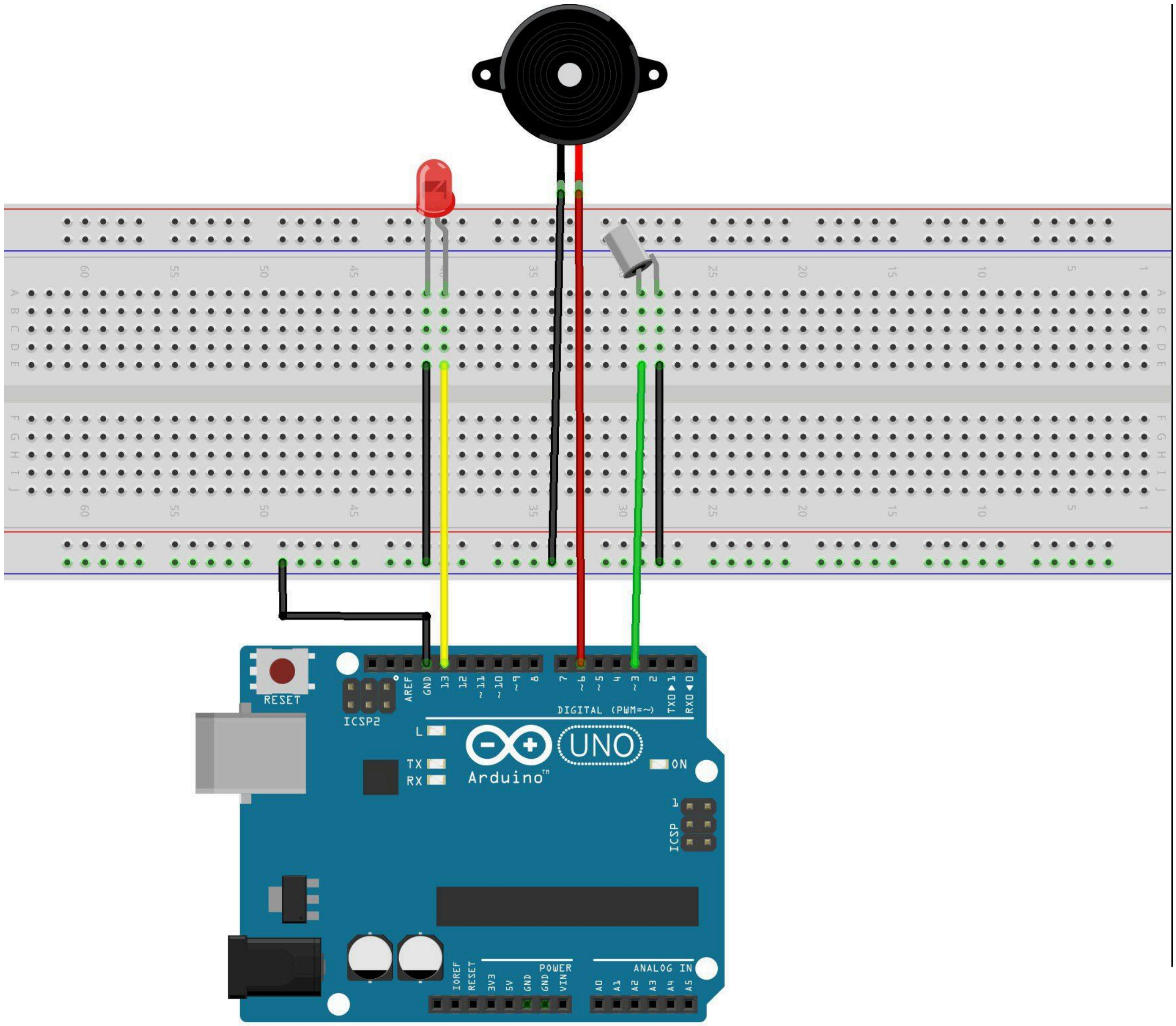
Procedure

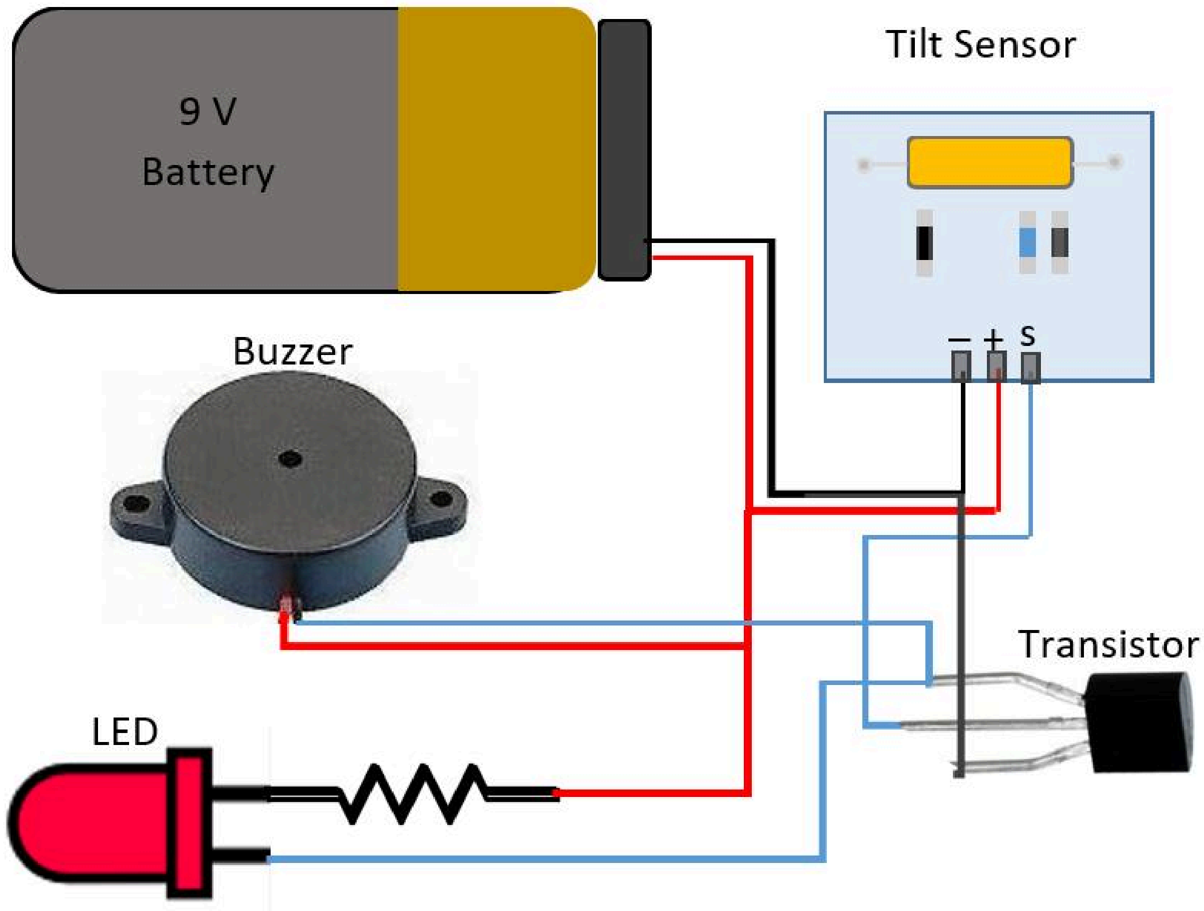
- Tap the table/sensor gently

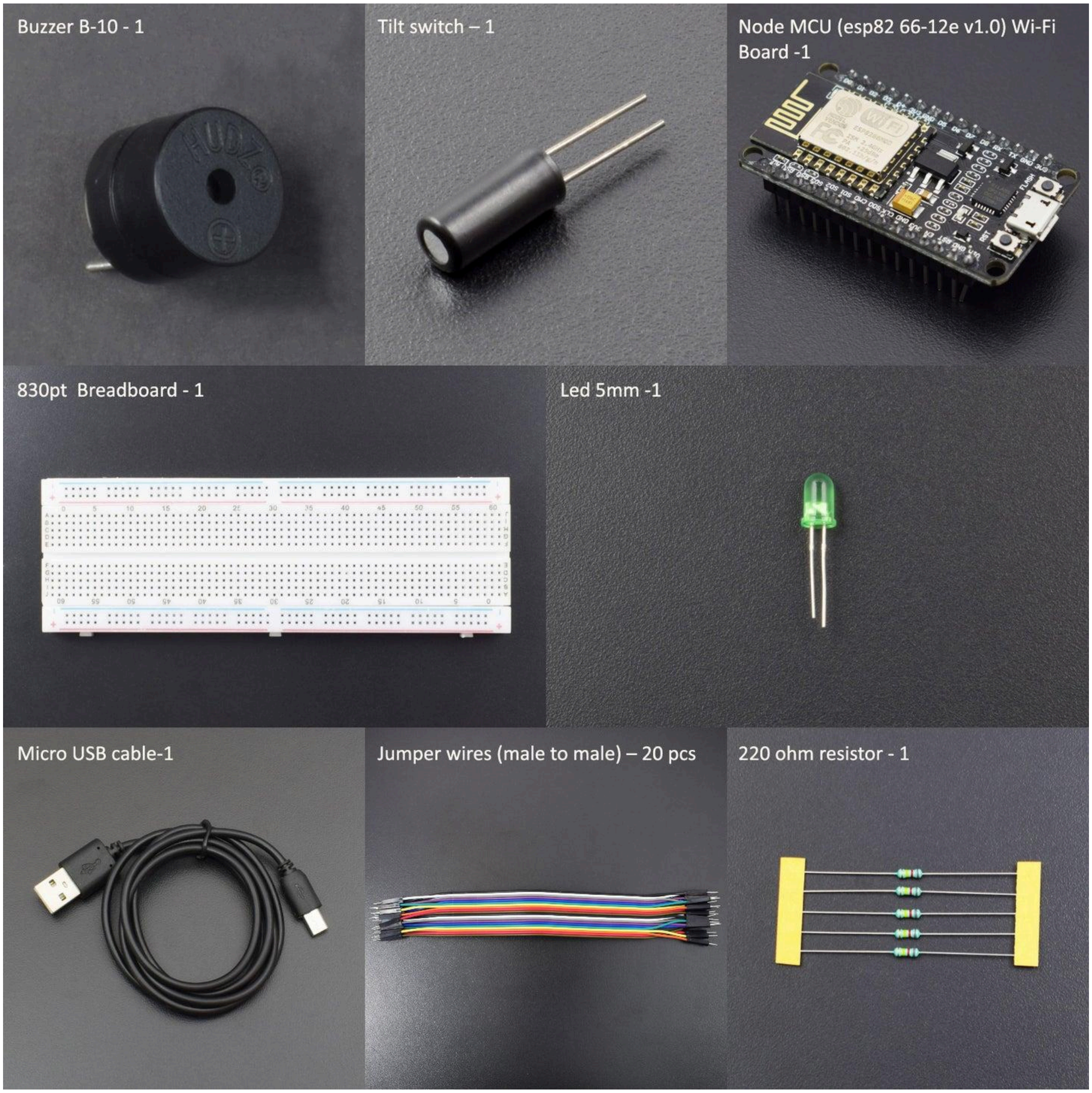
Expected Output

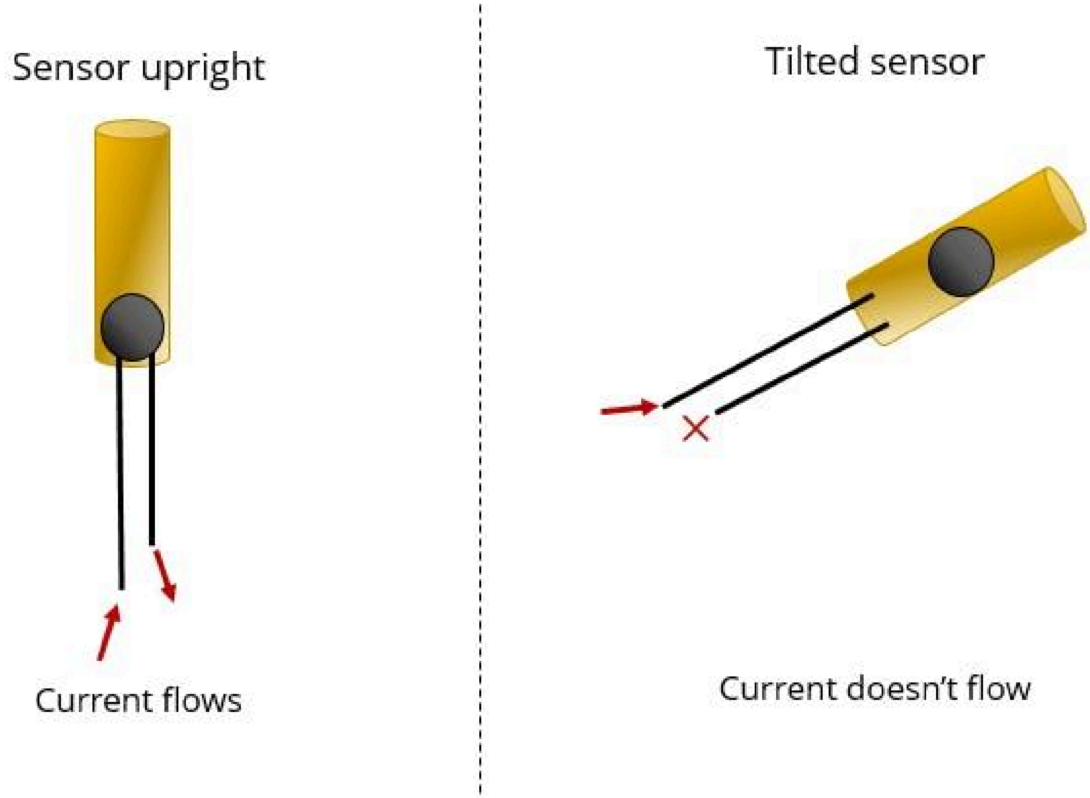
- Buzzer ON
- OLED shows alert
- Serial monitor logs alert

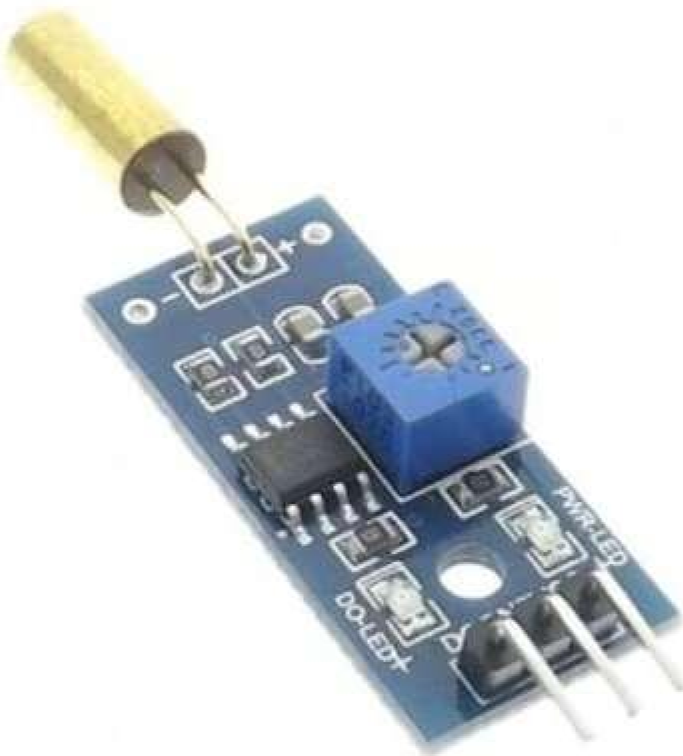
Test 3: Tilt Detection




9 V
Battery
Tilt Sensor
− + S
Buzzer
Transistor
LED


Buzzer B-10 - 1
Tilt switch – 1
Node MCU (esp82 66-12e v1.0) Wi-Fi Board -1
830pt Breadboard - 1
Led 5mm -1
Micro USB cable-1
Jumper wires (male to male) – 20 pcs
220 ohm resistor - 1


Sensor upright
Current flows
Tilted sensor
Current doesn't flow


PWR-LED
DO-LED

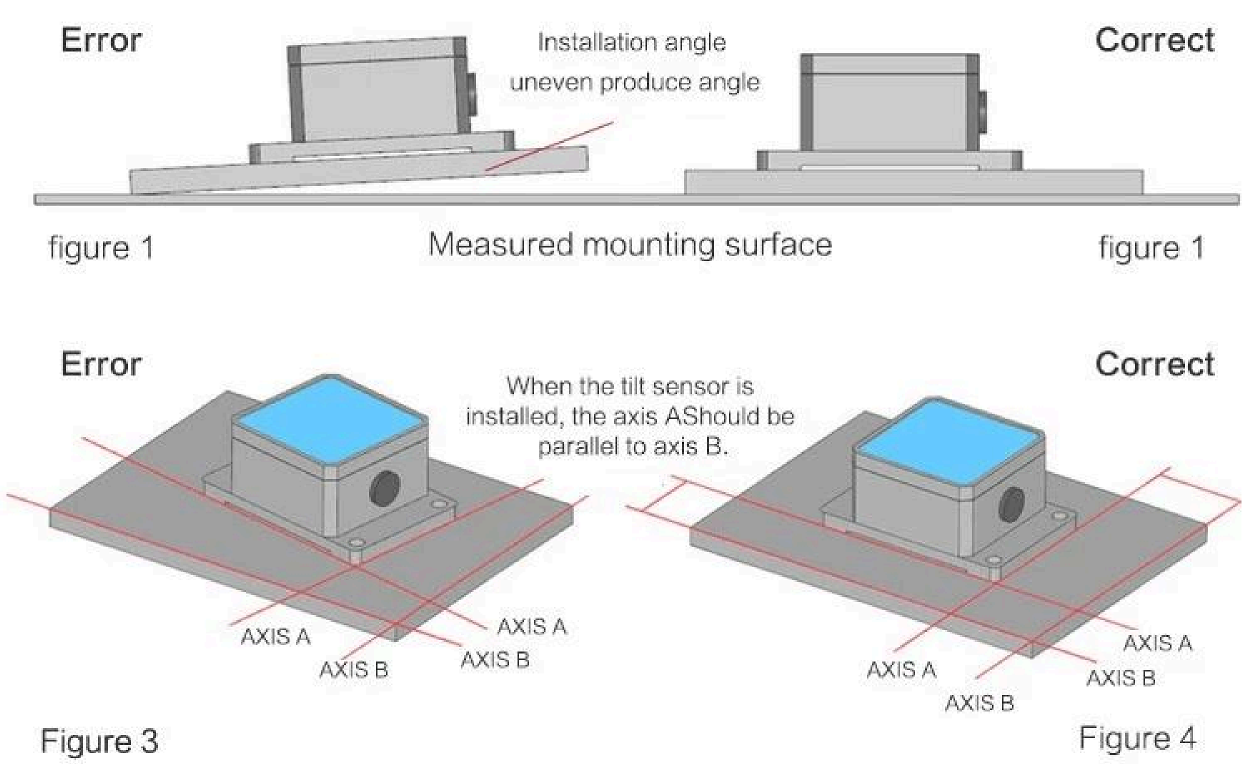
INSTALLATION PRECAUTIONS

Product installation precautions

Please follow the correct method to install the tilt sensor, improper installation will lead to measurement error, with particular attention to the side, the second line

1) The mounting surface of the sensor must be close to the surface to be measured, smooth and stable. If the installation surface is uneven, it is easy to cause the error of the sensor measurement.

2) the sensor axis and the measured axis must be parallel, the two axes as far as possible not to produce the angle, see Figure 3, 4



Procedure

- Tilt/lift the setup slightly

Expected Output

- Alarm triggers
- OLED alert shown

🔵 Test 4: Combined Attack

Procedure

- Shake + tilt simultaneously

Expected Output

- Strong alert
- Continuous buzzer

⚙️ 12. Sensitivity Adjustment

On SW-420 module:

- Rotate potentiometer

Clockwise

→ More sensitive

Anti-clockwise

→ Less sensitive

🎬 13. Demo Strategy

🧰 Setup

- Place system on table
- Keep OLED visible

🎤 Demo Flow

1. Show safe state
2. Tap sensor
3. Tilt setup
4. Explain theft simulation

🧠 14. Viva Explanation

> "The system uses vibration and tilt sensors to detect tampering and movement. The Arduino processes sensor data and triggers real-time visual and audio alerts."

🎥 15. Video References

📳 Vibration Sensor Tutorial

Watch Tutorial

📺 OLED Display Tutorial

Watch Tutorial

🚗 Anti-Theft System Demo

[Watch Demo]

🔥 16. Future Enhancements

- ESP32 mobile alerts
- GSM SMS alerts
- AI anomaly detection
- Cloud dashboard
- Mobile app integration

🏁 17. Final Summary

Feature	Description
Project Type	Embedded + IoT
Cost	₹150–₹250
Difficulty	Easy–Medium
Sensors Used	Vibration + Tilt
Outputs	OLED + Buzzer
Future Scope	High